

Optics

Optics $\rightarrow$ interference

$\downarrow$
wave nature (Huygen's)

$\Rightarrow$ light has dwellness $\rightarrow$ wave and particle

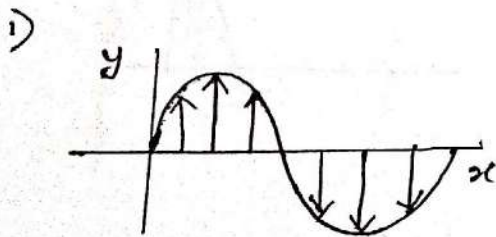
Wave :

Wave is one kind of disturbance that travels onward due to the repeated periodic motion of the particle

$\Rightarrow$ Waves are of two types

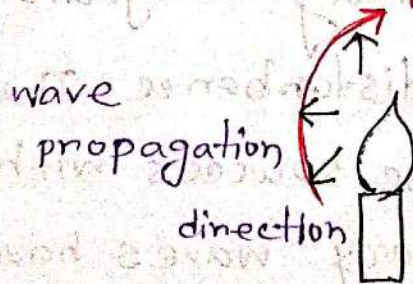
i) Transverse wave (আবর্ত তরঙ্গ)

ii) longitudinal wave (অনুদৈর্ঘ্য তরঙ্গ)
 $\downarrow$
sound wave



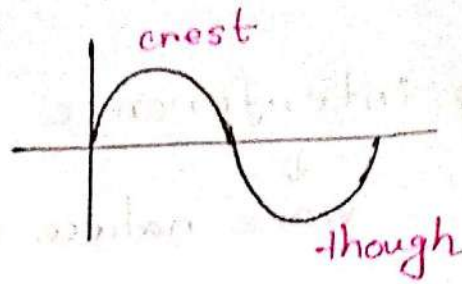
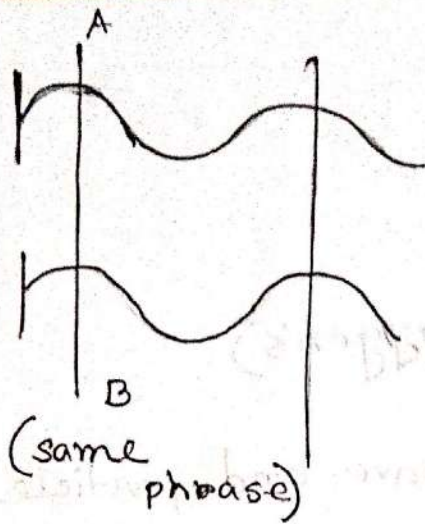
Transverse

primary wave form
(spherical wavefront)



$\rightarrow$ inward direction a spread

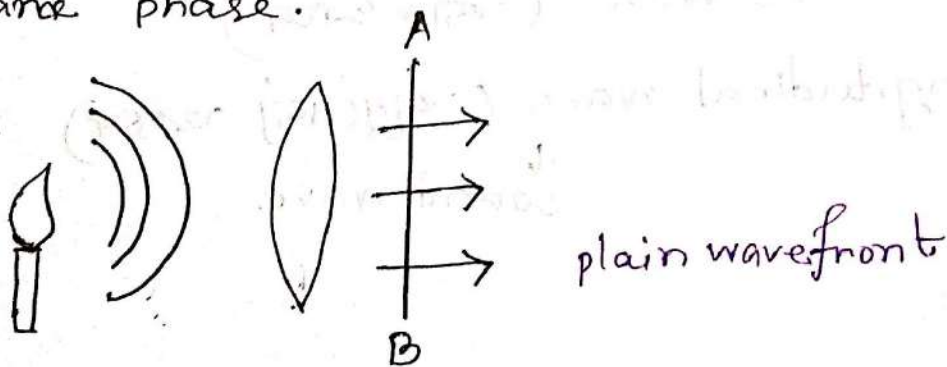
After an inward of time, the distance travelled the light wave from the source, $s = ct$



$AB = \text{wavefront (plane)}$

Wavefront :

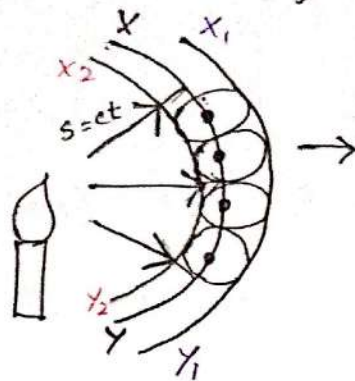
A wavefront is defined as the locus of all points which are in same state of vibration i.e. same phase.



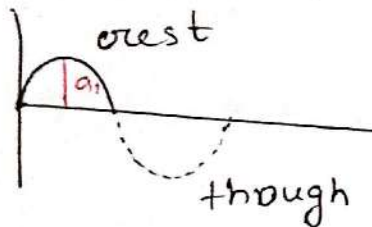
Huygen's Principle

1) Every point on a primary wavefront act as a secondary source of disturbance. The secondary waves spread from these sources with the same speed of as the primary waves have.

11) The surface that touches the secondary waves give rise the new position of new wavefront.



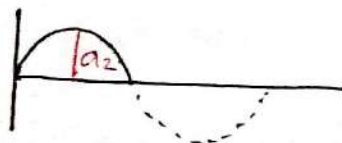
(Before superimposed)



$$y_1 = a_1 \sin \omega t$$

$$I_0 = a_1^2 = a^2$$

$$\delta = 2n\pi$$

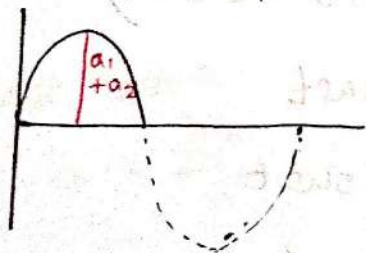


$$y_2 = a_2 \sin(\omega t + \delta)$$

$$A = (a_1 + a_2)$$

$$I_0 = a_2^2 = a^2$$

(After superimposed)



$$I = A^2$$

$$\therefore I = a^2 + a^2$$

$$= 2a^2$$

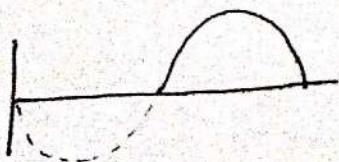
$$I = 4a^2$$

$$= 4I_0$$

$$\delta = (2n+1)\pi$$

$$A = 0$$

$$I = 0$$



u

$$I_{avg} = \frac{I_{max} + I_{min}}{2}$$

$$= \frac{4a^2 + 0}{2}$$

$$= 2a^2$$



Q

Interference obey the law of conservation energy, prove that.

#

$$y_1 = a \sin \omega t$$

$$y_2 = a \sin(\omega t + \delta)$$

$$y = y_1 + y_2$$

$$= a \sin \omega t + (a \sin \omega t \cos \delta + a \sin \delta \cos \omega t)$$

$$= a \sin \omega t (1 + \cos \delta) + a \sin \delta \cos \omega t$$

$$= \sin \omega t a (1 + \cos \delta) + a \sin \delta \cos \omega t$$

$$= \sin \omega t \times A \cos \theta + A \sin \theta \cos \omega t$$

$$= A \sin \omega t \cos \theta + A \cos \omega t \sin \theta$$

$$= A \sin(\omega t + \theta)$$

$$\left[\begin{array}{l} \text{suppose,} \\ A \cos \theta = a(1 + \cos \delta) \\ A \sin \theta = a \sin \delta \end{array} \right.$$



তারিখ:
পৃষ্ঠা :

$$A \cos \theta = a(1 + \cos \delta) \quad \text{--- ①}$$

$$A \sin \theta = a \sin \delta \quad \text{--- ②}$$

$$\text{①}^2 + \text{②}^2 \Rightarrow$$

$$A^2 = a^2 + 2a^2 \cos \delta + a^2 \cos^2 \delta + a^2 \sin^2 \delta$$

$$= 2a^2 + 2a^2 \cos \delta$$

$$= 2a^2 (1 + \cos \delta)$$

$$= 2a^2 \cdot 2 \cos^2 \frac{\delta}{2}$$

$$\Rightarrow A^2 = 4a^2 \cos^2 \frac{\delta}{2}$$

$$I = A^2$$

$$\Rightarrow I = 4a^2 \cos^2 \frac{\delta}{2}$$

$$\Rightarrow I = 4I_0 \cos^2 \left(\frac{\delta}{2} \right)$$

$$\Delta x = n\lambda$$

$$\Rightarrow \delta = \frac{2\pi}{\lambda} \times \Delta x$$

$$= \frac{2\pi}{\lambda} \times n\lambda$$

$$= 2n\pi$$

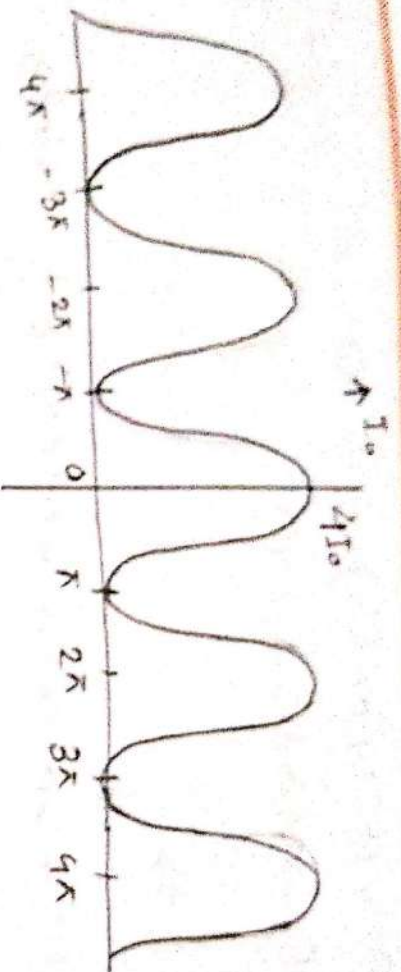
$$\Delta x = (2n+1) \frac{\lambda}{2}$$

$$\delta = \frac{2\pi}{\lambda} + (2n+1) \frac{\lambda}{2}$$

$$= (2n+1)\pi$$

$$I = 0$$

$$I = 4I_0$$



Condition of interference

i) light should be monochromatic
(one color light)

$5893 \text{ \AA}^\circ$ (Avg)

ii) coherent source

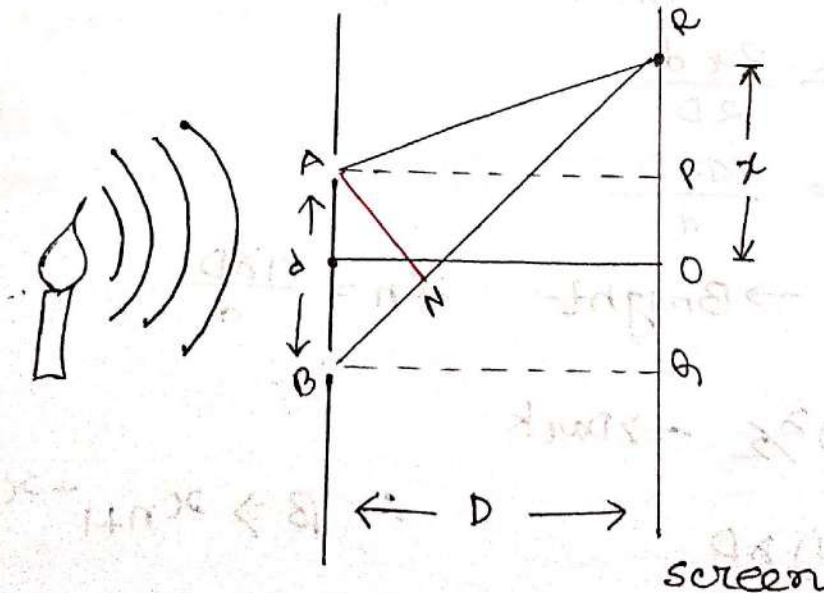
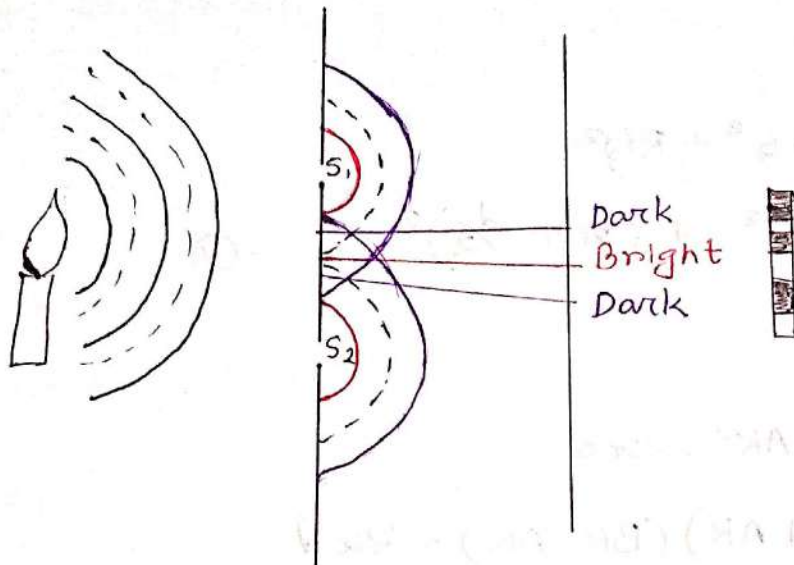
↓

same wavelength
same frequency (amplitude same)

Method : Division of waveform
Division of amplitude

Young's Double-slit Experiment

⇒ obeys huygen's law



$$AB = d = PQ$$

$$OP = OQ = d/2$$

Path difference,

$$4x = BR - AR$$

$\triangle APR \Rightarrow$

$$AR^2 = AP^2 + PR^2$$

$$= D^2 + \left(x - \frac{d}{2}\right)^2 \dots \dots \dots \textcircled{1}$$

$\triangle BQR \Rightarrow$

$$BR^2 = BQ^2 + RQ^2$$

$$= D^2 + \left(x + \frac{d}{2}\right)^2 \dots \dots \dots \textcircled{2}$$

$$\textcircled{2} - \textcircled{1} \Rightarrow$$

$$BR^2 - AR^2 = 2xd$$

$$\Rightarrow (BR + AR)(BR - AR) = 2xd$$

$$\Rightarrow BR - AR = \frac{2xd}{BR + AR}$$

$$\Rightarrow 4x = \frac{2xd}{2D}$$

$$\Rightarrow x = \frac{4xD}{d}$$

$$4x = n\lambda \rightarrow \text{Bright} \quad x_n = \frac{n\lambda D}{d}$$

$$4x = (2n+1)\frac{\lambda}{2} \rightarrow \text{Dark}$$

$$x_n = \frac{(2n+1)\lambda D}{2d}$$

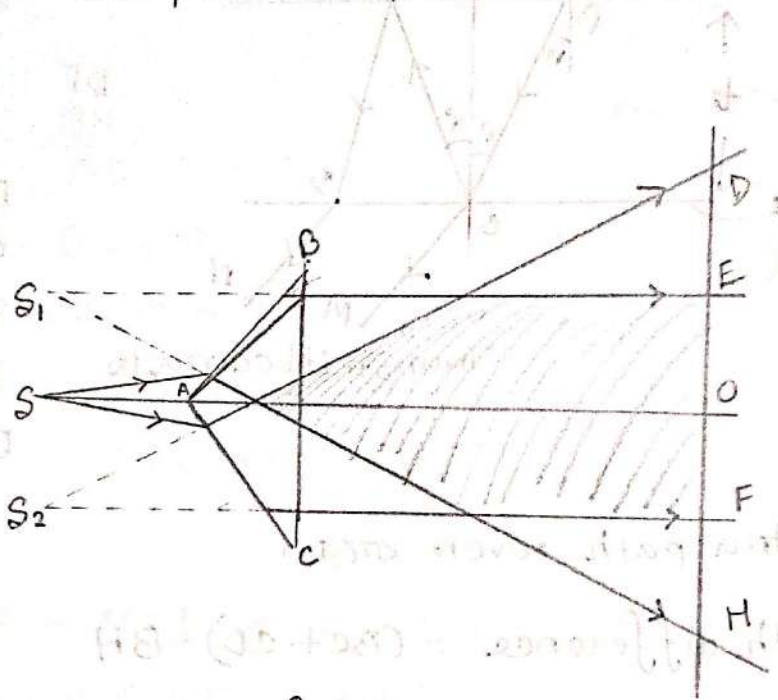
$$x_{n+1} = \frac{(2n+3)\lambda D}{2d}$$

$$\therefore \beta \Rightarrow x_{n+1} - x_n = \frac{\lambda D}{d}$$

Condition of good interference fringe:

$$\beta = \frac{\lambda D}{d}$$

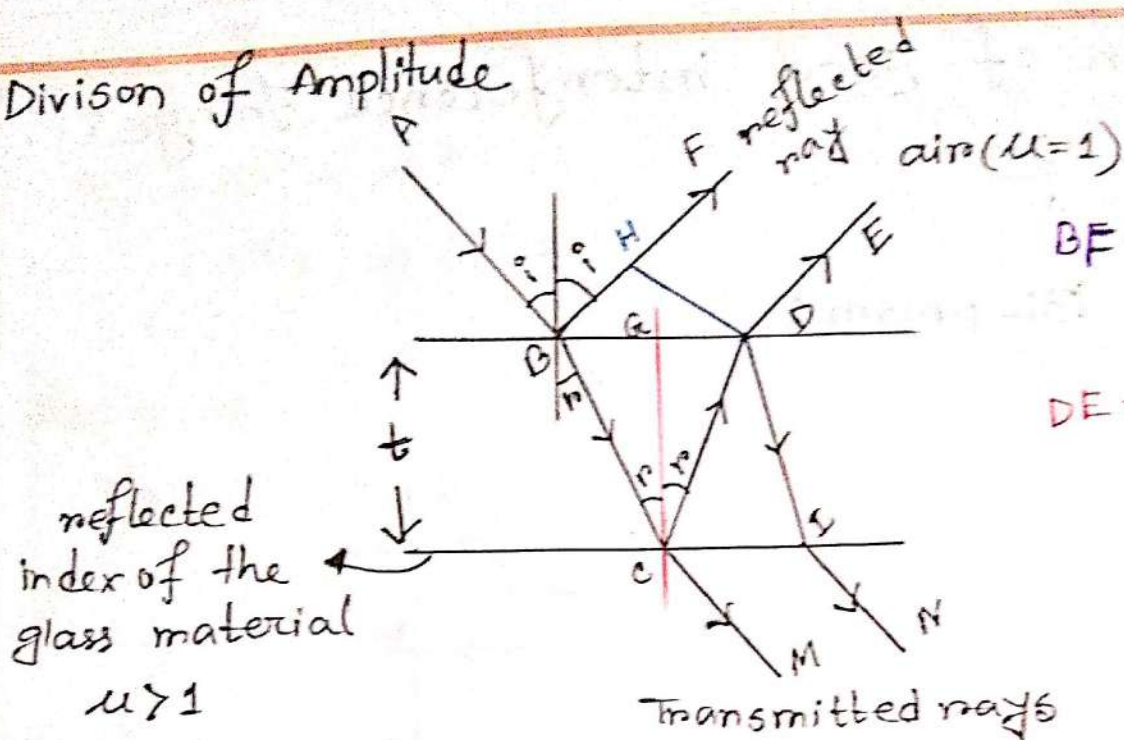
Fresnel Bi-prism:



Thin Film Interference

Thin film is a layer of transparent material bounded by two surface having thickness.

Division of Amplitude



$BF = DE =$
coherent source

$$DE = HF$$

$$BC = CD$$

$$CG = t$$

$$BD = BG + GD$$

$$BG = GD$$

$$BD = 2BG$$

$(BC + CD)$ extra path covered by ray

$$\therefore BD \text{ is Path difference} = (BC + CD) - BH$$

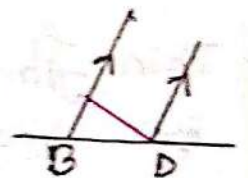
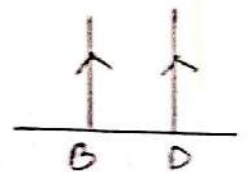
$$\Delta x = (BC + CD) - BH$$

Optical path = Geometrical Path $\times \mu$

$$\Delta x = \underbrace{\mu}_{\substack{\downarrow \\ \text{glass} \\ \text{material}}} (BC + CD) - BH \rightarrow \text{air medium } \mu = 1$$

$$= 2\mu BC - BH \text{ ----- (1)}$$

$$BC = CD$$



$\triangle BCG,$

$$\cos r = \frac{CG}{BC}$$

$$\Rightarrow BC = \frac{t}{\cos r}$$

$\triangle BCG,$

$$\tan r = \frac{BG}{CG}$$

$$\Rightarrow BG = t \times \tan r$$

$\triangle BDH,$

$$\sin i = \frac{BH}{BD}$$

$$\Rightarrow BH = BD \times \sin i$$

$$\Rightarrow BH = 2BG \times \sin i$$

$$= 2t \tan r \times \sin i$$

$$= 2\mu t \sin r \times \tan r$$

we know,

$$\mu = \frac{\sin i}{\sin r}$$

$$\sin i = \mu \times \sin r$$

$$\therefore \Delta x = \mu (BC + CD) - BH$$

$$= 2\mu BC - BH$$

$$= \frac{2\mu t}{\cos r} - 2\mu t \sin r \tan r$$

$$= \frac{2\mu t}{\cos r} - \frac{2\mu t \sin^2 r}{\cos r}$$

$$= \frac{2\mu t}{\cos r} (1 - \sin^2 r)$$

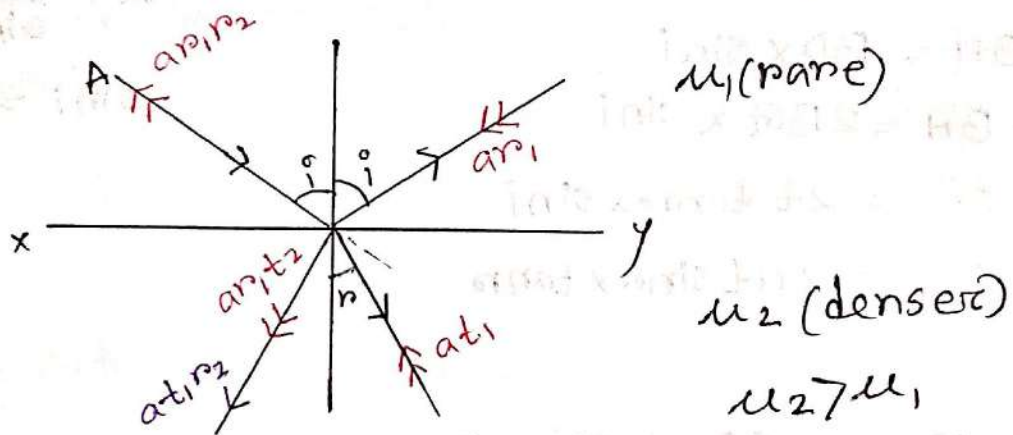
$$= \frac{2\mu t}{\cos r} \times \cos^2 r$$

$$\Delta x = 2\mu t \cos r$$

$$\Delta x = 2u + \cos r \pm \boxed{\frac{\lambda}{2}}$$

Phase change on reflection

→ Stokes theorem



$$ar_1 t_2 + at_1 r_2 = 0$$

$$r_1 = -r_2$$

$$\phi = \pi$$

t_1, t_2 denser medium
 t_1 & t_2 almost same

r_1 = rare medium

r_2 = denser medium

$$\phi = \frac{2\pi}{\lambda} \times \Delta x$$

$$\Rightarrow \Delta x = \frac{\lambda}{2\pi} \times \phi$$

$$\Rightarrow \Delta x = \frac{\lambda}{2\pi} \times \pi$$

$$= \frac{\lambda}{2}$$

Phase change 180° কারণ
(negative) sign আছে।

$$\Delta x = 2ut \cos r + \frac{\lambda}{2}$$

$$\Delta x = n\lambda \rightarrow \text{Bright}$$

$$2ut \cos r + \frac{\lambda}{2} = n\lambda$$

$$\Rightarrow 2ut \cos r = (2n-1) \frac{\lambda}{2} \rightarrow \text{Bright}$$

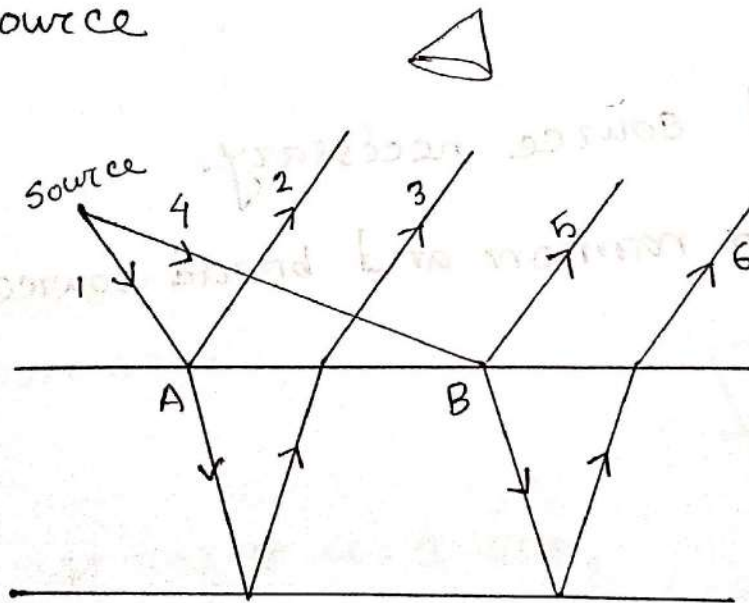
$$\Delta x = (2n+1) \frac{\lambda}{2} \rightarrow \text{dark}$$

$$\Rightarrow 2ut \cos r + \frac{\lambda}{2} = (2n+1) \frac{\lambda}{2}$$

$$\Rightarrow 2ut \cos r = n\lambda + \frac{\lambda}{2} - \frac{\lambda}{2}$$

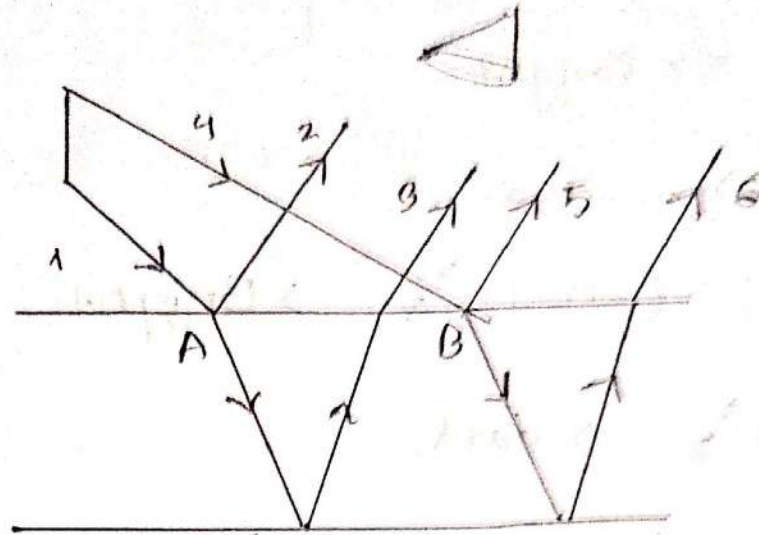
$$\Rightarrow 2ut \cos r = n\lambda \rightarrow \text{Dark}$$

Narrow Source



Narrow Source থেকে thin film এ আলো পড়লে একটি portion দেখা যায় না, limit হয়ে থাকে।

Broad Source



বড় portion হুঁড়ে interference pattern তৈরি
পারবে।

⇒ Thin film এর জন্য Broad source use করা।

Q

Broad source necessary.

↳ Draw narrow and broad source.

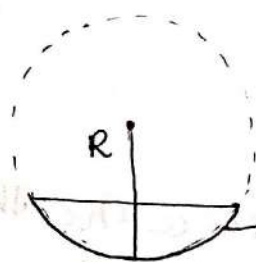
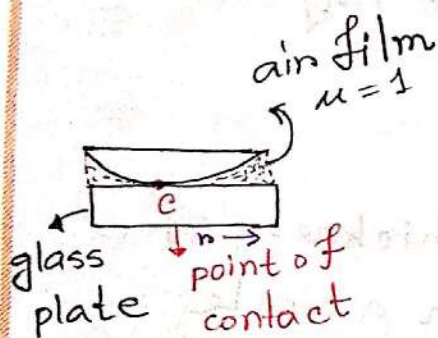
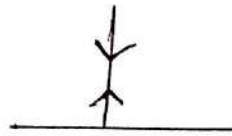
Newton's Ring

- i) Formation of Newton's Ring.
- ii) Why Newton's are circular in shape?
- iii) Application

At normal incident

$$r = 0$$

$$\cos r = 1$$



plano convex lens

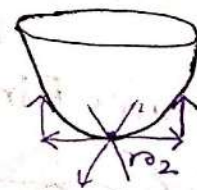
$R \rightarrow$ radius of curvature

$$2\mu t \cos r = n\lambda$$

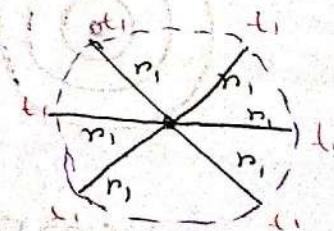
↑
thickness

1

convex lens এর ক্ষেত্রে $\mu = 1$ হবে,
কারণ মাঝে air film থাকে।



যদিও
thickness
সমান

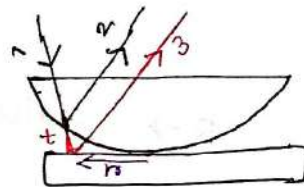
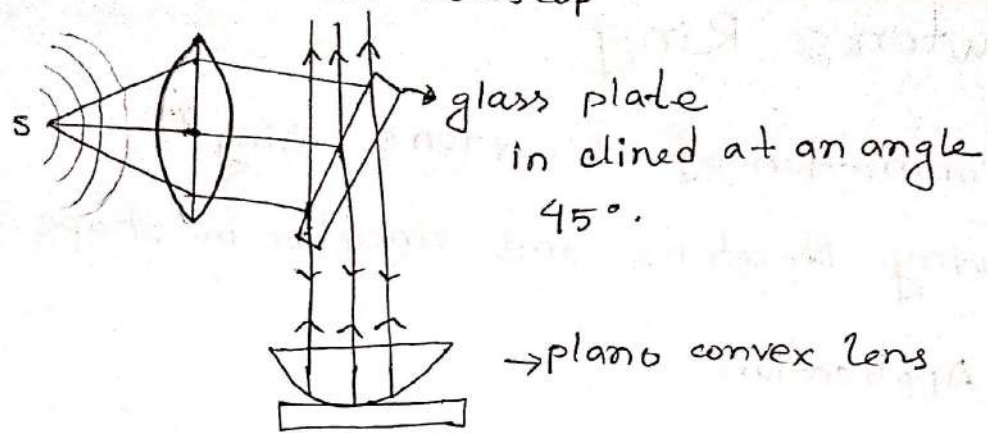


interference pattern
circular.

→ Newton's ring

তারিখ: _____
পৃষ্ঠা: _____

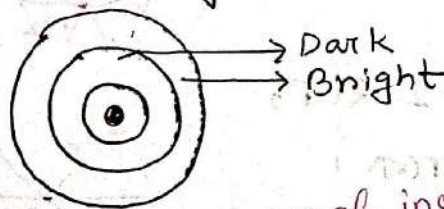
Telescope



Divison of amplitude
যেকো → একাট ন্যু যেকো
২ টি ন্যু তৈরি।

⇒ The locus of points where the thickness of air film is same that fall on a circle about the point of contact.

Newton's Ring



→ normal incident

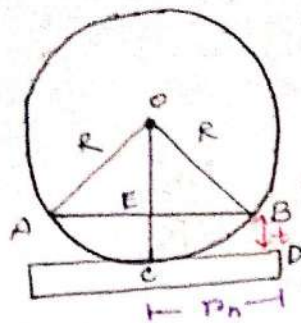
$$2\mu t \cos r = n\lambda \rightarrow \text{Dark}$$

$$2t = n\lambda \quad \text{--- ①}$$



তারিখ: _____

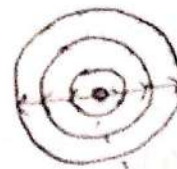
পৃষ্ঠা: _____



$$BD = CE = t$$

$$CD = BE = r_n$$

$$OE = R - t$$



$$D_n = 2R_n$$

$$\Rightarrow R_n^2 = \frac{D_n^2}{4}$$

400E ,

$$OB^2 = OE^2 + BE^2$$

$$\Rightarrow R^2 = (R - t)^2 + r_n^2$$

$$\Rightarrow R^2 = R^2 - 2Rt + t^2 + r_n^2$$

$$\Rightarrow r_n^2 = 2Rt - t^2$$

$$\Rightarrow r_n^2 = 2Rt$$

$$\Rightarrow 2t = \frac{r_n^2}{R}$$

$$\Rightarrow 2t = \frac{D_n^2}{4R}$$

D_{n+p}

↓ ring number
no difference

→ + এর মান খুবই ছোট, তাই
 t^2 এর মান খুবই ক্ষণিক

① এর মান বাড়িয়ে ,

$$n\lambda = \frac{D_n^2}{4R}$$

$$\Rightarrow D_n^2 = 4n\lambda R \text{ --- (i)}$$

③ - ② ⇒

$$D_{n+p}^2 - D_n^2 = 4p\lambda R$$

$p \rightarrow$ ring difference

যদি air medium না থাকে,

$$2\mu t \cos r = n\lambda$$

$$\Rightarrow 2\mu t = n\lambda$$

$$\Rightarrow 2t = \frac{n\lambda}{\mu}$$

$$\therefore 2 \cdot l = \frac{D_n^2}{4R}$$

$$\therefore \frac{D_n^2}{4R} = \frac{n\lambda}{\mu}$$

$$\Rightarrow D_n^2 = \frac{4n\lambda R}{\mu}$$

$$\Rightarrow D_{n+p}^2 = \frac{4(n+p)\lambda R}{\mu}$$

$$\left[D_{n+p}^2 - D_n^2 \right]_{\text{liq}} = \frac{\left[D_{n+p}^2 - D_n^2 \right]_{\text{air}}}{\mu}$$

$$\Rightarrow \mu = \frac{\left[D_{n+p}^2 - D_n^2 \right]_{\text{air}}}{\left[D_{n+p}^2 - D_n^2 \right]_{\text{liquid}}}$$

Newton's Ring Center

$$\Delta x = 2\mu t \cos r + \lambda/2$$

$$= 2t + \lambda/2$$

$$= \lambda/2$$

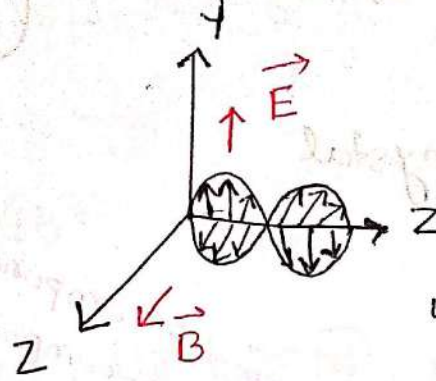
$$\delta = \frac{2\pi}{\lambda} \times \Delta x$$

$$= \frac{2\pi}{\lambda} \times \lambda/2$$

$$= \pi \rightarrow \text{destruction interference}$$

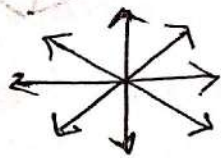
অত্যা newton ring এর center dark হয়।

Polarization of light

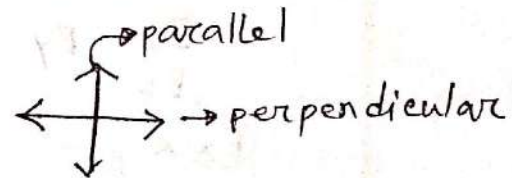


wave Propagation direction

light এর vibration $\rightarrow$ electric field এর vibration (non magnetic)



symmetrical

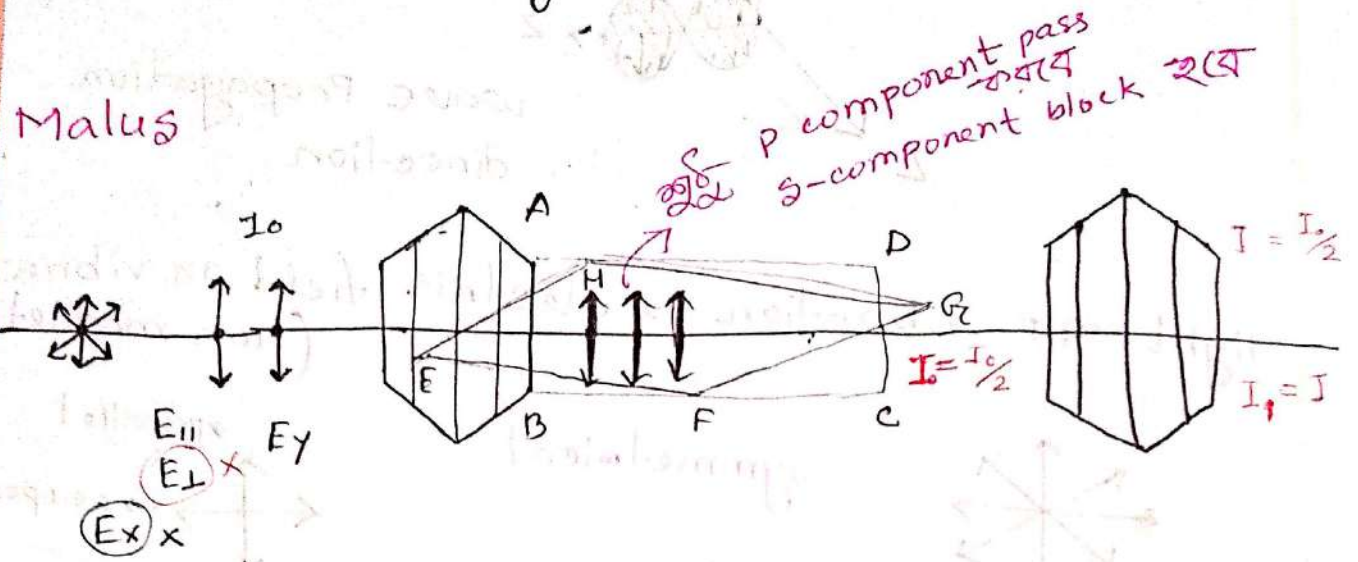
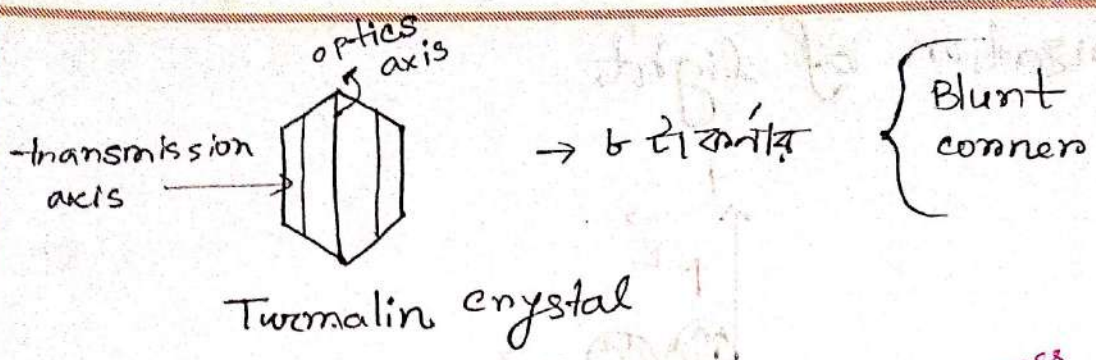


$\vec{E}_{||}$
Board এর parallel
এর vibrate
(double headed arrow)

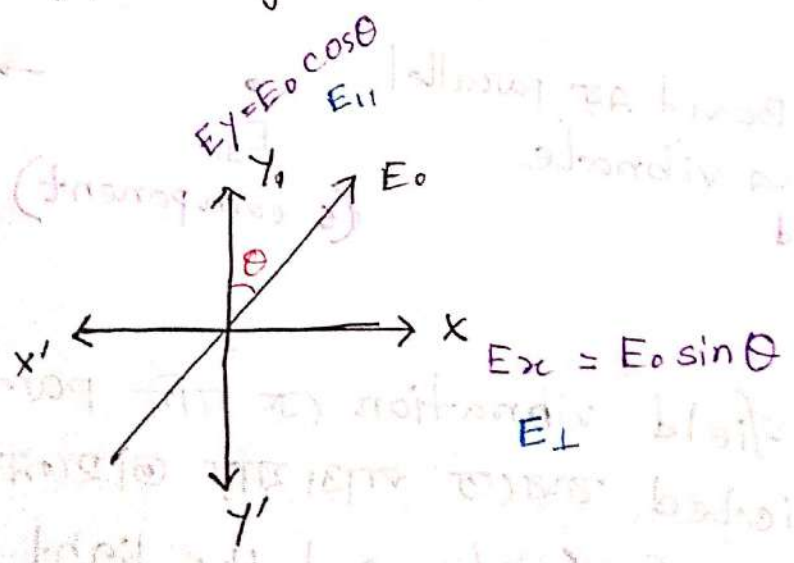
$E_{\perp}$
(s component)
perpendicular

$\Rightarrow$ Electric field vibration কে যদি particular plane এ restricted করতে পারা যায় তাহলে তাকে polarization of light. And the light is polarized light.

$\Rightarrow$ polar light এর intensity কমে যাবে/জার্ক হবে যাবে।



ABCD $\rightarrow$ plane of vibration



$$I_0 = E_0^2$$

$$I \propto E_y^2$$

$$I = E_0^2 \cos^2 \theta$$

$$I = I_0 \cos^2 \theta$$

analyzer $\leftarrow \therefore I_1 = I \cos^2 \theta \rightarrow$ polarized light intensity
 $\downarrow$
 polarizer

$$I_1 = I \cos^2 \theta$$

$$\Rightarrow \frac{1}{2} I_0 = I_0 \cos^2 \theta$$

$$\Rightarrow \cos \theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = 45^\circ$$

$\Rightarrow$ Unpolarized light যদি polarizer এর মতো দিয়ে যায় এবং Intensity হবে $I = \frac{I_0}{2} \cos^2 \theta$

$\Rightarrow$ Polarizer থেকে বের হয়ে analyzer এর মতো ৯ দিয়ে দ্রবেশ করলে $I_1 = I \cos^2 \theta$ (polarizer এর respect এ analyzer)

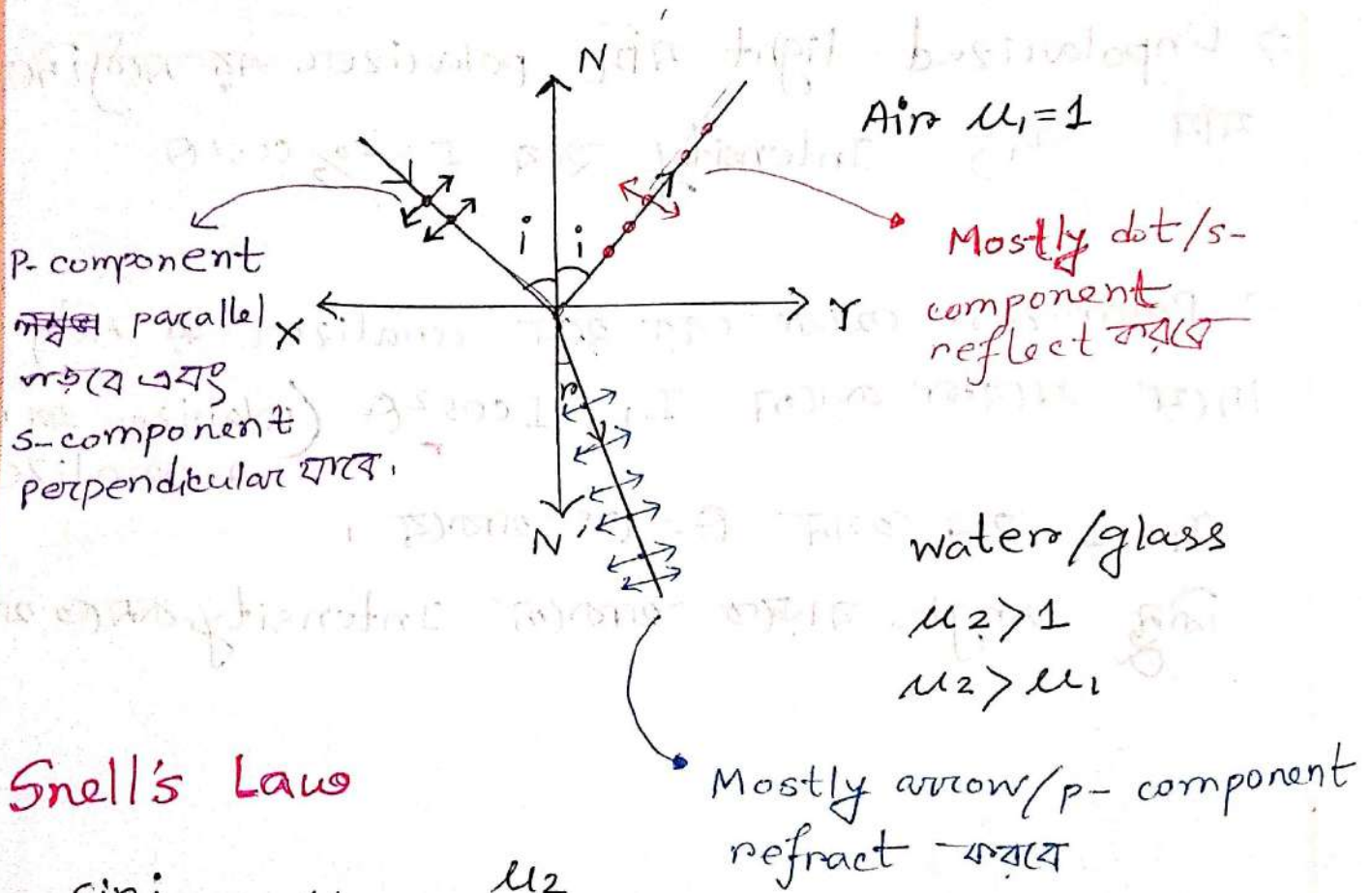
$I_1 = I$ হবে তখন $\theta = 0^\circ$ থাকবে।

কিন্তু angle বাড়তে থাকলে Intensity কমে থাকবে।

$$\frac{I_1}{I_0} = \cos^2 \theta = \frac{I_1}{I_0}$$

Unpolarized light ko polarized করার নিয়ম :

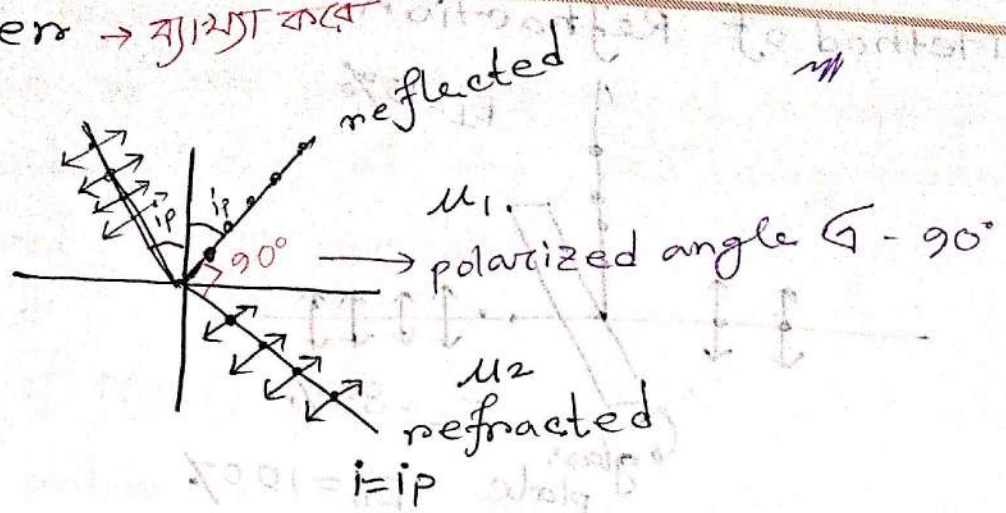
- i) by-reflection
- ii) by-refraction
- iii) double refraction



Snell's Law

$$\frac{\sin i}{\sin r} = \mu_2 = \frac{\mu_2}{\mu_1}$$

Brewster $\rightarrow$ স্রোতা কণ



$$\tan i_p = \frac{\mu_2}{\mu_1}$$

$$i_p = \tan^{-1}(\mu_2)$$

water = 1.33

$$i_p = 53^\circ$$

$\rightarrow$ Angle of polarization

glass = 1.52

$$i_p = \tan^{-1}(\mu_2)$$

$$i_p = 56^\circ$$

$$\tan i_p = \frac{\sin i_p}{\sin r}$$

$$\Rightarrow \frac{\sin i_p}{\cos i_p} = \frac{\sin i_p}{\sin r}$$

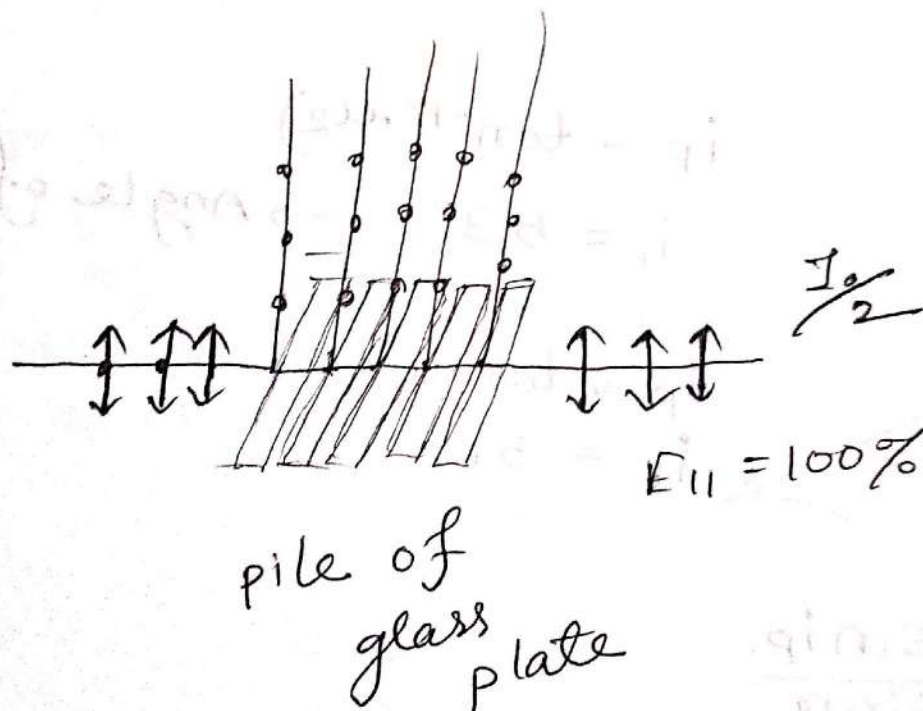
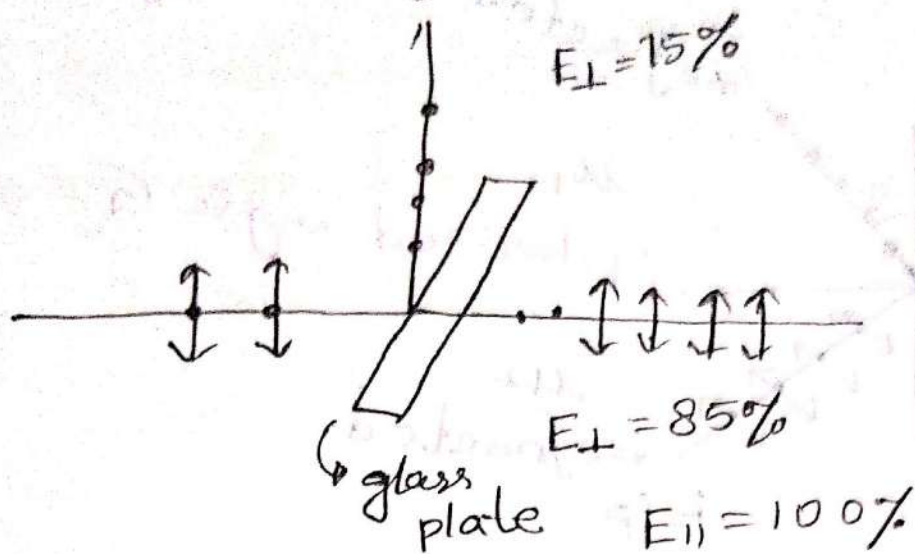
$$\Rightarrow \cos i_p = \sin r$$

$$\Rightarrow \sin(90^\circ - i_p) = \sin r$$

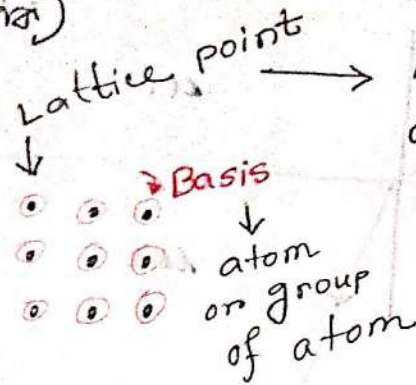
$$\Rightarrow 90^\circ - i_p = r$$

$$\Rightarrow i_p + r = 90^\circ$$

Method of Refraction



Calcite Crystal $\rightarrow$ Rhombohedral
(আনাইসোট্রপিক)



A regular and periodic arrangement of points in space.

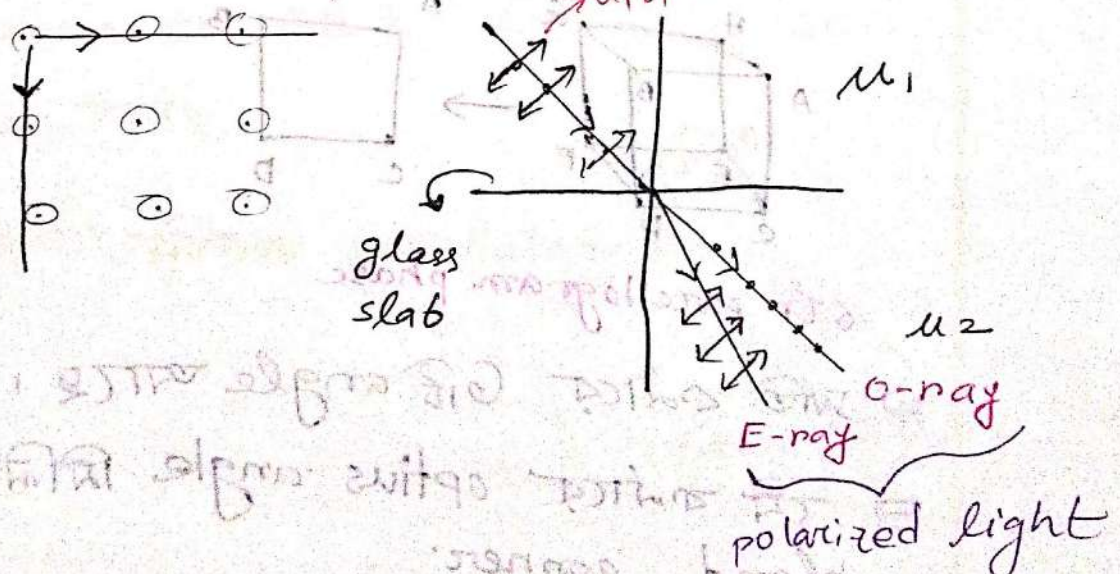
Lattice + Basis = Crystal structure

Crystal structure কে তিনভাবে ভাগ করা যায়।

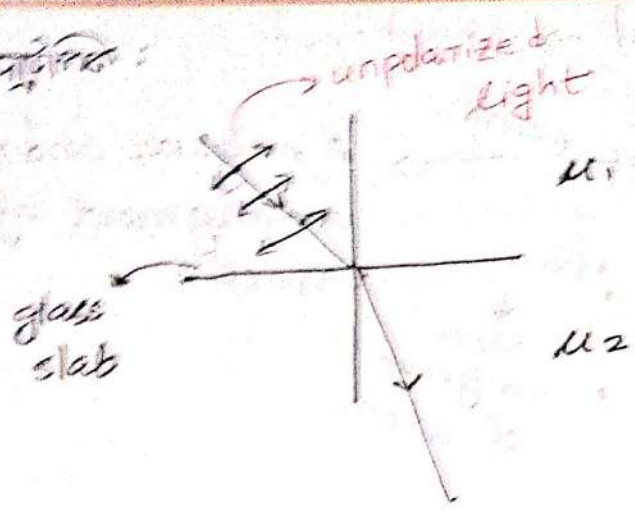
- i) crystalline solid
- ii) non-crystalline solid
- iii) polycrystalline

আইসোট্রপিক $\rightarrow$ independent, glass slab

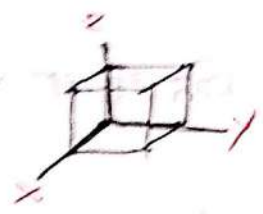
আনাইসোট্রপিক $\rightarrow$ direction wise, dependent



অন্যসংক্রিয়তা:



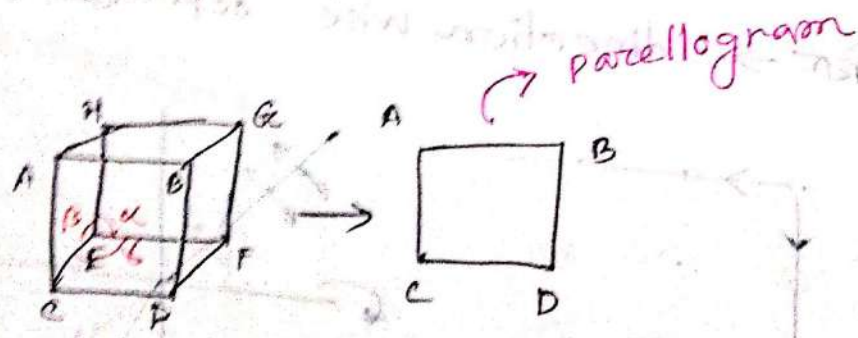
Crystal এর Fundamental box $\rightarrow$ Unit cell



calcite crystal

$\hookrightarrow$ dehydrated calcium-carbonate

$\hookrightarrow$ Iceland spar



৬টি parallelogram phase

$\Rightarrow$ প্রতি ক্রান্তের ৬টি angle আছে।

$\Rightarrow$ যে ক্রান্তের obtuse angle মিলিত হয় তাকে blunt corner

AEFH - principle plane

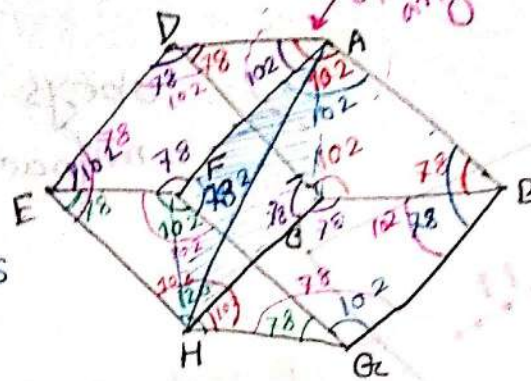
optical angle

Acute angle $\rightarrow < 90^\circ$

optical angle $\rightarrow > 90^\circ$

AH - optic axis

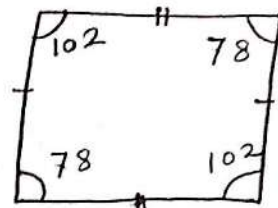
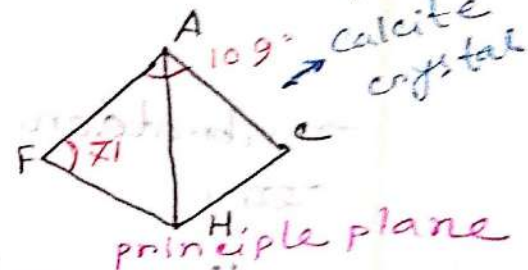
blond কক্ষ $\rightarrow A$ ও H
Blunt



$\Rightarrow A$ ও H এ-তিনটি 102 ডিগ্রি মিলিত হবে.

জানকি (জানকি)

Date: 02-12-2025

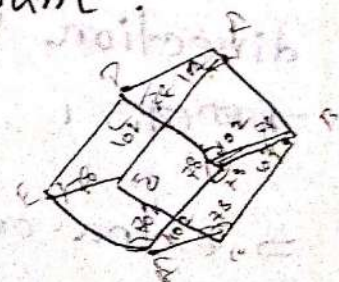


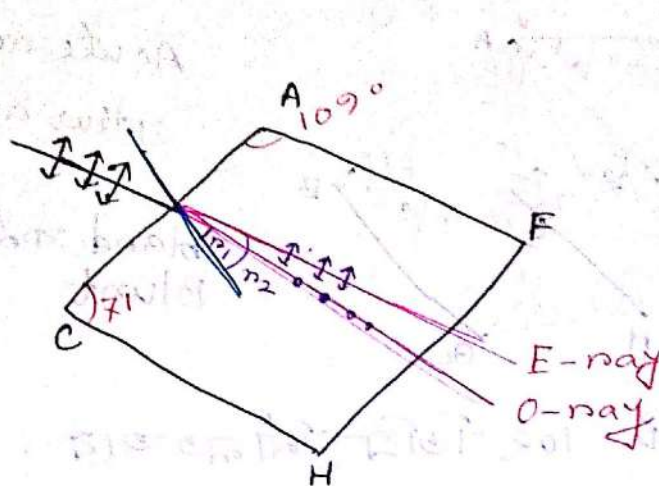
$$\alpha + \beta = 180$$

$\Rightarrow$ crystal এ optic axis দুই একটা থাকবে
সব Uni-axis crystal.

$\Rightarrow$ principle plane এর ABGF ও CDEH এর
perpendicular আছে।

$\Rightarrow$ principle plane নিজেও parallelogram.





obeys law of refraction

⇒ vibration এক জনের সাথে অন্যর perpendicular হয়।

⇒ linear polarized

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \frac{\mu_2}{\mu_1} = \frac{c_o}{c_{in}}$$

velocity of light in air medium

velocity of light in water / glass medium

⇒ Ordinary ray calcite crystal এক মের direction এ যাক না কেন velocity same থাকবে।

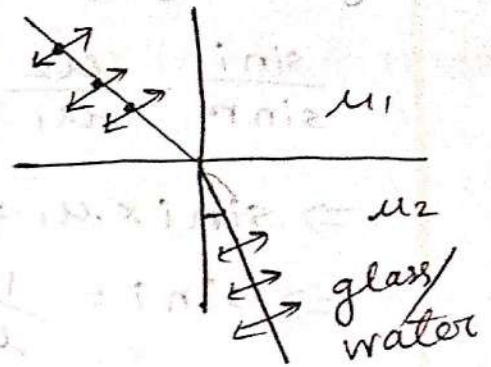
⇒ extra-ordinary ray → মেনে চলে না।

$$\mu_o = 1.658$$

$$\mu_e = 1.486$$

$$\mu_o > \mu_e$$

$$\mu = \frac{c_o}{c_{in}}$$

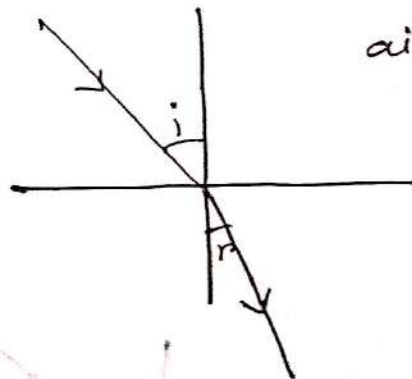
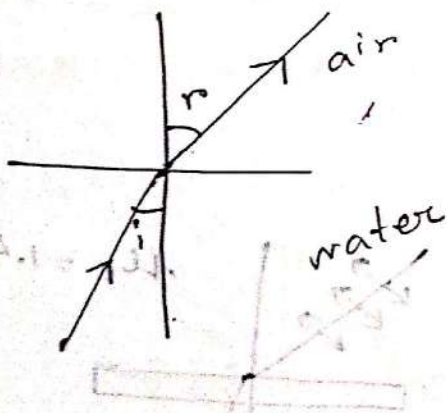


$\Rightarrow$ e-ray থেকে fast যাবে।

TIR

$\hookrightarrow$ Total Internal reflection

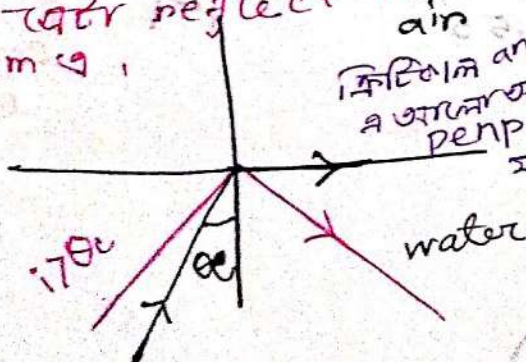
dense $\rightarrow$ less ~~water~~ air



air $\mu_1 = 1$
 $\mu_2 = 1$

water/glass
 $\mu_2 > 1$
 $\mu_1 > 1$

$\Rightarrow$ critical angle এর থেকে বেশি হলে তেঁর reflect করে denser medium এ।



critical angle
অন্যভাবে
perpendicular
যাবে।

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \sin i \times \mu_1 = \sin r \mu_2$$

$$\Rightarrow i = \sin^{-1}(\mu_2)$$

By changing, $\mu_1 \rightarrow \mu_2$
 $\mu_2 \rightarrow \mu_1$

$$i = \sin^{-1}\left(\frac{1}{\mu_1}\right)$$

$\theta_c =$ critical angle

By changing,

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

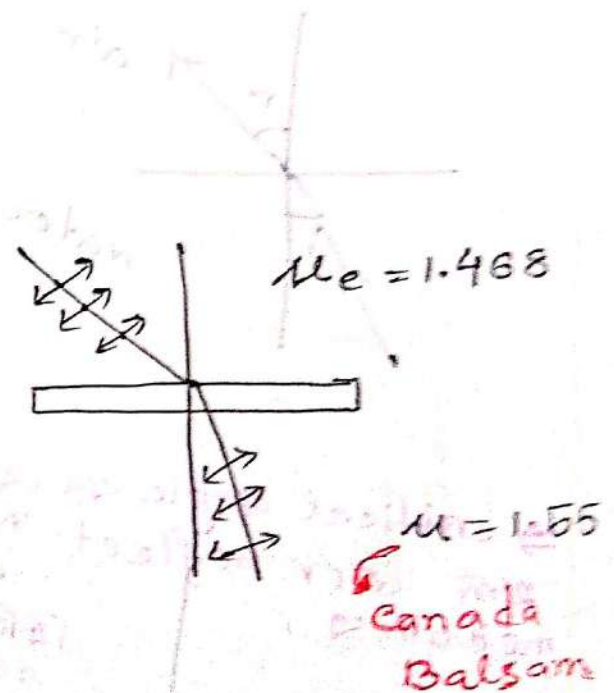
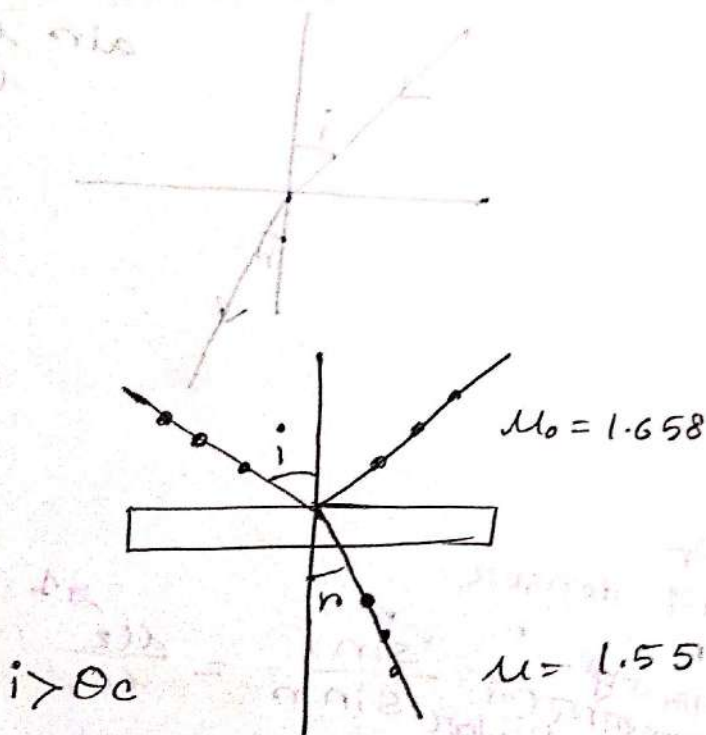
$$\Rightarrow \sin i \times \mu_1 = \sin r \times \mu_2$$

$$\Rightarrow \sin i = \frac{1}{\mu_1}$$

$$\Rightarrow i = \sin^{-1}\left(\frac{1}{\mu_1}\right)$$

$$\Rightarrow \theta_c = i = \sin^{-1}\left(\frac{1}{\mu_1}\right)$$

$$\Rightarrow \mu_o = 1.658, \mu_e = 1.486 \text{ (निकान)} \quad \text{AIT}$$



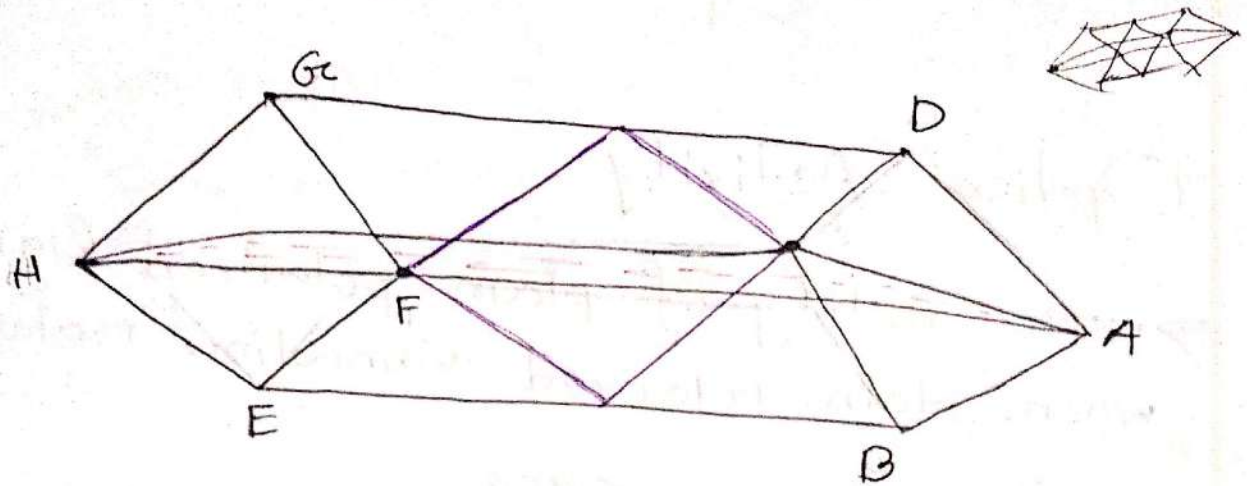
critical angle,

$$\theta_c = \sin^{-1}\left(\frac{\mu_2}{\mu_1}\right)$$

$$= \sin^{-1}\left(\frac{1.55}{1.658}\right) = 69^\circ$$

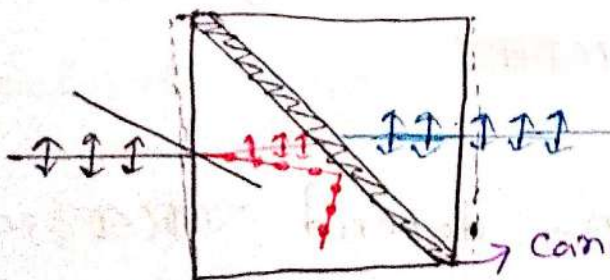
Nicole Prism

$\Rightarrow$ its length is 3 times greater than its width



$\Rightarrow$ Nicole prism এর plane কে এমনভাবে ঘর্ষা হয়েছে যেন Angle 21° থাকে যার কোণ 68° হয়।

polarizer হিসেবে কাজ করে



Canada Balsam $\rightarrow$ glue হিসেবে

Q Explain the construction of ~~conce~~ nicole prism.

- ⇒ $\hookrightarrow$ rotation parameter is α
clockwise / anticlockwise
- ⇒ Lovenice half-shade polarimeter

Optical Activity

⇒ The ability of plane polarized light when plane polarized vibration rotate

⇒ ক্রাইস্টালিটি show করে

⇒ clockwise $\rightarrow$ dextro rotatory
ঘনমিত $\rightarrow$ দক্ষিণ ঘূর্ণন - ডাক্তার

⇒ Anticlockwise $\rightarrow$ levo rotatory
বামমিত $\rightarrow$ বাম ঘূর্ণন - কোকেন

⇒ Angle of rotation depend করে concentration
এবং চৈদ্র, depend on thickness (solution),
depend on solution nature

$$\theta \propto c$$

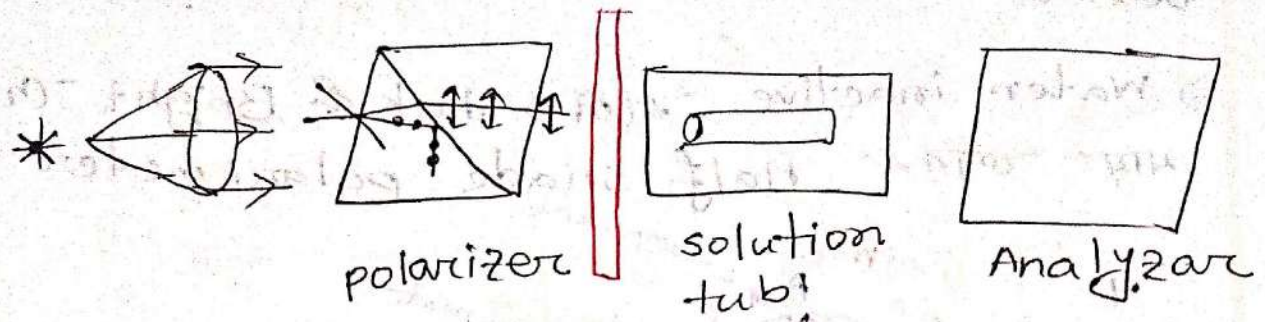
$$\theta \propto l$$

$$\therefore s = \frac{\theta}{lc}$$

$$\Rightarrow \theta \propto lc$$

$$\Rightarrow \theta = slc$$

$$s = \frac{100\theta}{lc}$$

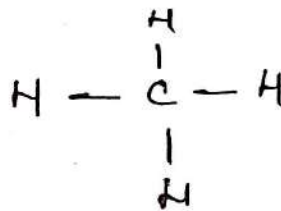
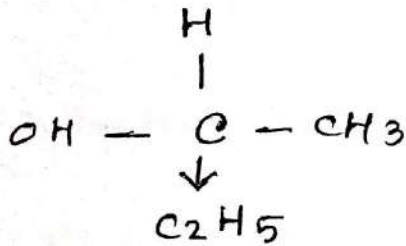


water $\rightarrow$ optically inactive



#

Date: 07-012-25



করাযানিচি আছে $\rightarrow$ থাকেনই Polarized হবে.

$\Rightarrow$ Optical activity নির্ভর করে temperature, c, l, nature.

$$\theta \propto c$$

$$\theta \propto l$$

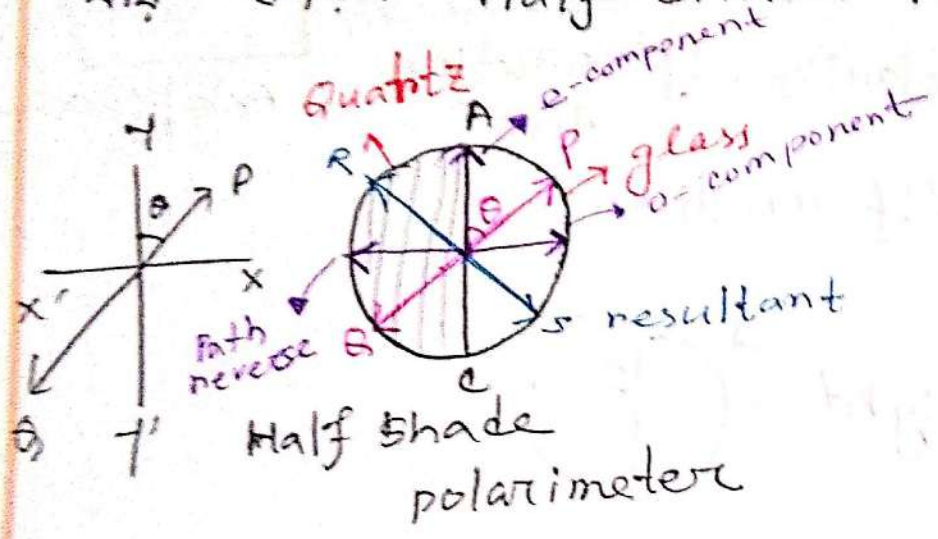
$$\theta \propto lc$$

$$\therefore \theta = slc$$

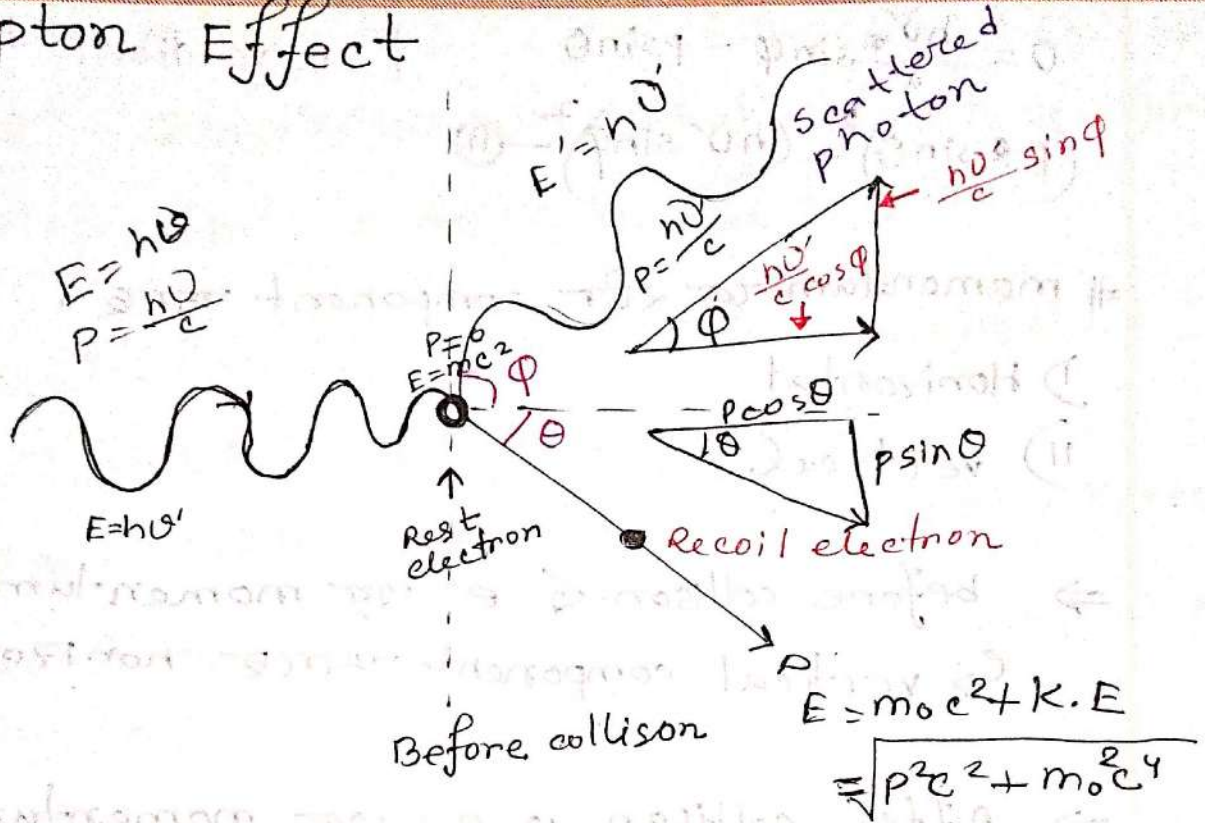
$$\Rightarrow \boxed{s = \frac{100\theta}{lc} \text{ (cm}^{-1}\text{)}} \quad (\text{cm}^{-1})$$

Dextro

⇒ Water inactive, dark & Bright then
Half shade polarimeter



Compton Effect



$$(h\nu - h\nu') = K.E$$

⇒ After collision এর wavelength বড় হবে।

⇒ After and before collision energy & momentum conserve হয়ে থাকবে!

$$\frac{h\nu}{c} + 0 = \frac{h\nu'}{c} \cos \phi + p \cos \theta \quad [\text{In horizontal direction}]$$

$$\Rightarrow \frac{h\nu}{c} - \frac{h\nu'}{c} \cos \phi = p \cos \theta$$

$$\Rightarrow (h\nu - h\nu' \cos \phi) = (pc \cos \theta)^2 \quad \text{--- (1)}$$

$$0 = \frac{h\nu'}{a} \sin \phi - p \sin \theta$$

[In vertical direction]

$$(p \sin \theta)^2 = \left(\frac{h\nu'}{a} \sin \phi \right)^2 \quad \text{--- (11)}$$

momentum has 2 components

i) Horizontal

ii) vertical

⇒ before collision e^- has momentum $h\nu$ horizontal.

⇒ vertical component is zero.

⇒ After collision e^- has momentum $h\nu'$ at an angle θ to the horizontal.

$$\text{(1) + (11)} \Rightarrow$$

$$p^2 c^2 = (h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu') \cos \phi$$

$$E = m_0 c^2 + (h\nu - h\nu')$$

$$E = (p^2 c^2 + m_0^2 c^4)^{1/2}$$

$$(p^2 c^2 + m_0^2 c^4)^{1/2} = m_0 c^2 + (h\nu - h\nu')$$

$$\Rightarrow p^2 c^2 + m_0^2 c^4 = \{m_0 c^2 + (h\nu - h\nu')\}^2$$

$$\Rightarrow p^2 c^2 + m_0^2 c^4 = m_0^2 c^4 + 2m_0 c^2 (h\nu - h\nu') + (h\nu - h\nu')^2$$

$$\Rightarrow p^2 c^2 = 2m_0 c^2 (h\nu - h\nu') + (h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu')$$

$$\Rightarrow p^2 c^2 = (h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu') \cos \phi$$

$$2m_0 c^2 (h\nu - h\nu') + (h\nu)^2 + (h\nu')^2 + 2(h\nu)(h\nu') =$$

$$(h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu') \cos \phi$$

$$\Rightarrow 2m_0 c^2 (h\nu - h\nu') - 2(h\nu)(h\nu') = -2(h\nu)(h\nu') \cos \phi$$

$$\Rightarrow 2(h\nu)(h\nu') - 2(h\nu)(h\nu') \cos \phi = 2m_0 c^2 (h\nu - h\nu')$$

$$\Rightarrow 2(h\nu)(h\nu') [1 - \cos \phi] = 2m_0 c^2 (h\nu - h\nu')$$

$$\Rightarrow h(\nu\nu') (1 - \cos \phi) = m_0 c^2 (\nu - \nu')$$

$$\Rightarrow \frac{h}{m_0 c^2} (1 - \cos \phi) = \frac{\nu - \nu'}{\nu\nu'}$$

$$\Rightarrow h c^2 \frac{1}{\lambda\lambda'} (1 - \cos \phi) = m_0 c^3 \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right)$$

$$\Rightarrow h \frac{1}{\lambda\lambda'} (1 - \cos \phi) = m_0 c \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right)$$

$$\Rightarrow \frac{h}{m_0 c} (1 - \cos \phi) = \lambda\lambda' \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right)$$

$$\Rightarrow \frac{h}{m_0 c} (1 - \cos \phi) = \lambda' - \lambda$$

$$\Rightarrow \lambda' = \lambda + \frac{h}{m_0 c} (1 - \cos \phi)$$

$$\Rightarrow \boxed{\lambda' = \lambda + \frac{h}{m_0 c} (1 - \cos \phi)} \quad \lambda' > \lambda$$

$$\frac{h}{m_0 c} = 2.43 \times 10^{-12}$$

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos\phi)$$

$$\phi = 180^\circ$$

$$\Delta\lambda = 0.048 \text{ \AA}$$

$$\lambda_{\max} = \sim 0.05 \text{ \AA}$$

% of change, % of change

$$= \frac{\Delta\lambda_{\max}}{\lambda_{\text{xray}}} = \frac{\Delta\lambda_{\max}}{\lambda_{\text{visible}}}$$

$$= \frac{0.05}{5}$$

$$= 0.01$$

$$= \frac{0.05}{5000}$$

$$= 10^{-5}$$

$$\# \frac{h}{m_0 c^2} (1 - \cos\phi) = \frac{U - U'}{U U'}$$

$$KE = hU - hU'$$

$$KE = E - E'$$

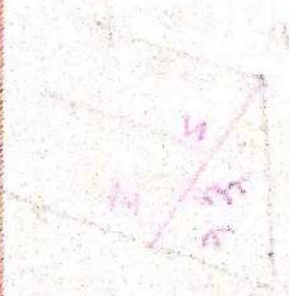
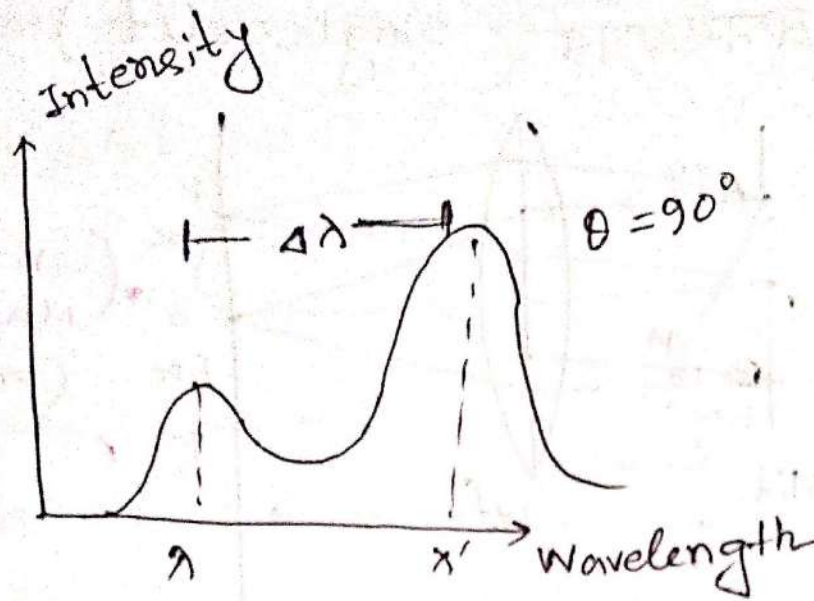
$$= hU - hU'$$

$$\Rightarrow \frac{h}{m_0 c^2} (1 - \cos\phi) = \frac{1}{U'} - \frac{1}{U}$$

$$\Rightarrow \frac{1}{U'} = \frac{1}{U} + \frac{h}{m_0 c^2} (1 - \cos\phi)$$

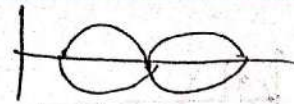
$$\Rightarrow U' = \frac{1}{\frac{1}{U} + \frac{h}{m_0 c^2} (1 - \cos\phi)}$$

$$hU' = E'$$



phase difference = 180°

Resultant = 0



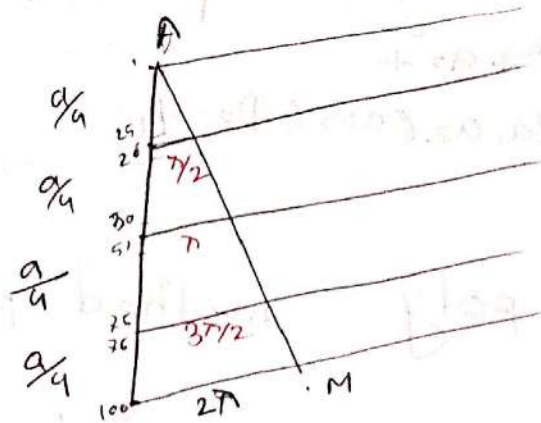
$$A = 0$$

$$I = A^2 = 0$$

51 ও 76 বরা মার্গ-
phase difference,

$$\frac{3\lambda}{2} - \lambda$$

$$\Rightarrow \frac{\lambda}{2}$$



$$a \sin \theta = \lambda$$

$$a \sin \theta = 2\lambda$$

$$\boxed{a \sin \theta = n\lambda} \rightarrow \text{secondary minima}$$

$$\text{Seco} \quad \boxed{a \sin \theta = (2n+1)\frac{\lambda}{2}} \rightarrow \text{secondary maxima}$$



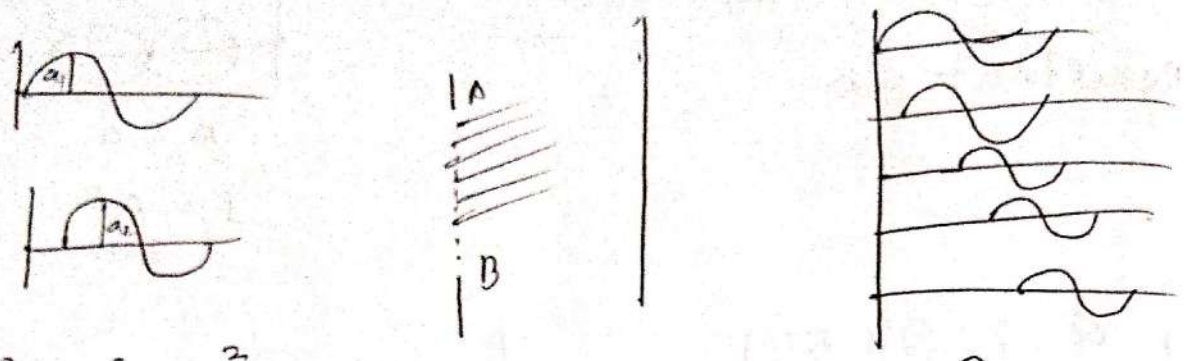
$$\tan \theta = \sin \theta = \frac{x}{f} = \frac{OP}{OC}$$

$$\Rightarrow \sin \theta = \frac{x}{f}$$

$$\frac{\lambda}{a} = \frac{x}{f}$$

$$\Rightarrow x = \frac{\lambda f}{a}$$

$$\therefore \text{width, } w = \frac{2\lambda f}{a}$$

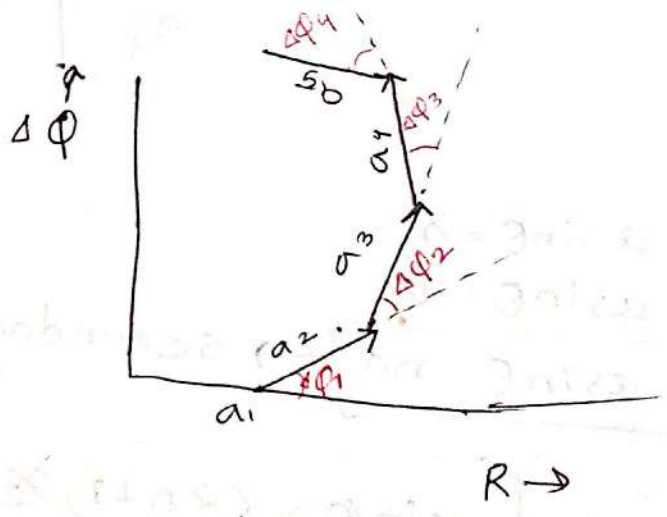
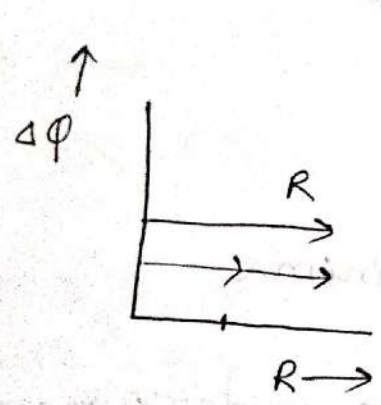


$$A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos(\phi_2 - \phi_1)$$

$$\Delta x = a \sin \theta$$

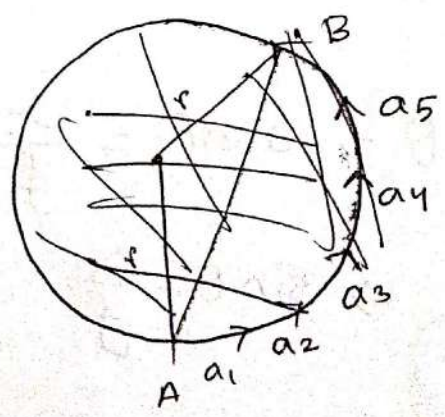
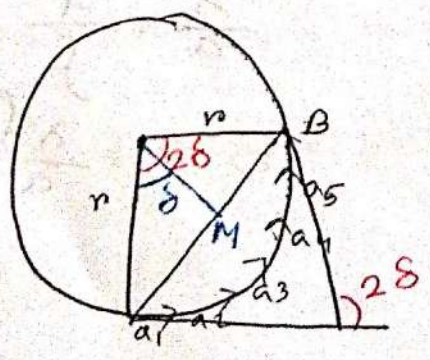
$$\Rightarrow \Delta \phi = \frac{2\pi}{\lambda} x \sin \theta$$

Vector poly method phasor diagram.



Circle ২৬

Resultant amplitude
= AB



$$AB = AM + BM$$

$$= 2AM$$

$$AM = BM$$

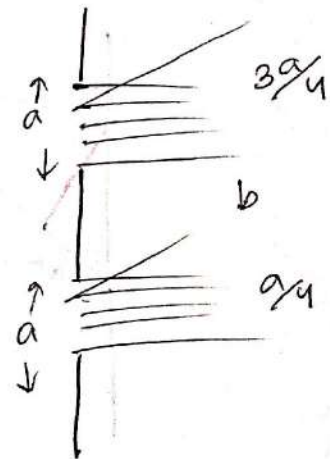
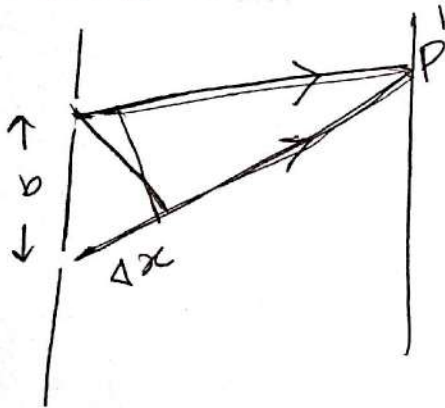
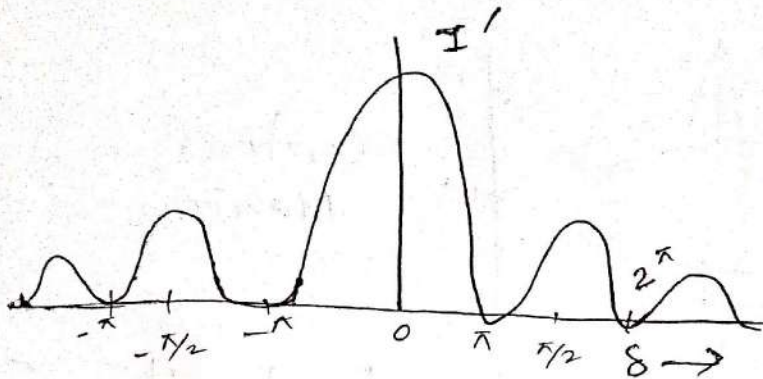
$$\sin \delta = \frac{AM}{AO}$$

$$AM = r \sin \delta$$

$$\therefore R = AB$$

$$= 2AM$$

$$R = AB = 2r \sin \delta$$



$$\Delta x = (a+b) \sin \theta = m\lambda$$

→ maxima

$$\Delta x = (a+b) \sin \theta = (2m+1) \frac{\lambda}{2}$$

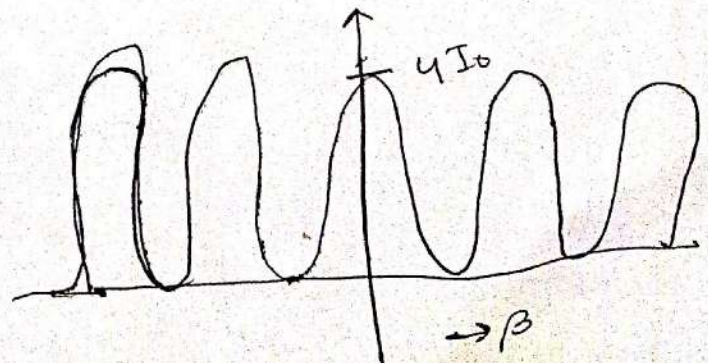
→ minima

$$2\beta = \frac{2\pi}{\lambda} \times (a+b) \sin \theta$$

$$\beta = \frac{\pi}{\lambda} (a+b) \sin \theta$$

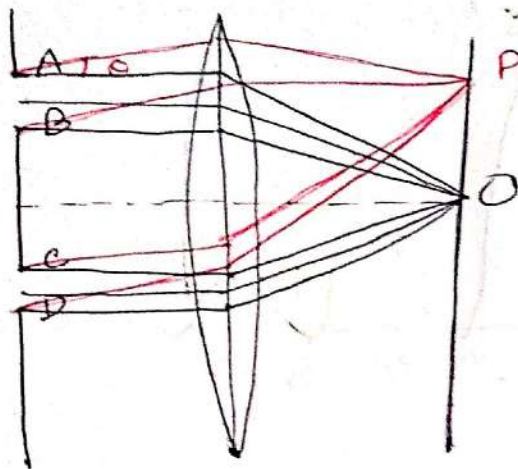
$$I = 4I_0 \cos^2 \frac{\delta}{2}$$

$$I = 4I_0 \cos^2 \beta$$



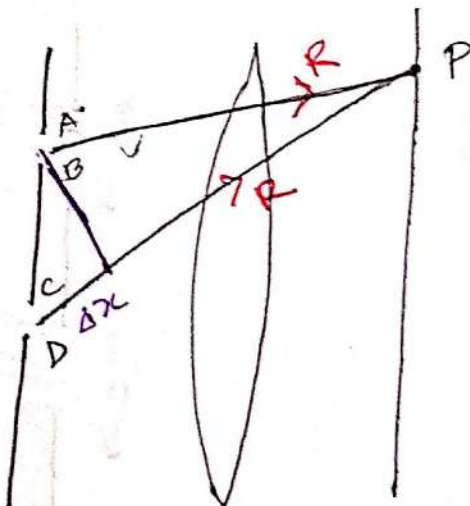
Resultant $\rightarrow R$

θ to bend



Central
Maxima

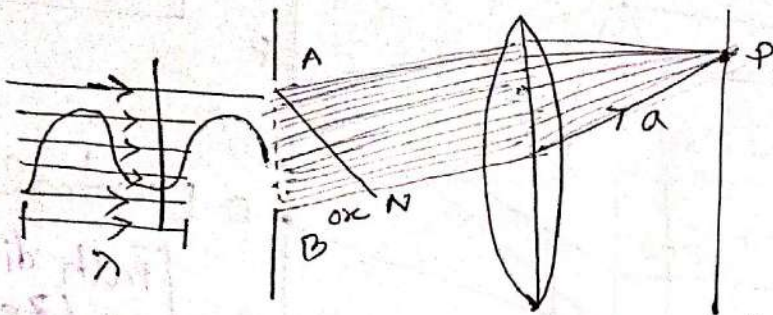
[Intensity ଦିଶି ଶାସ୍ତ୍ର]



$$\Delta x = (A+B) \sin \theta$$

$$\Delta x = a \sin \theta$$

$$R = (Ka) \frac{\sin \theta}{\theta}$$



$$\lambda \propto a$$

$$\lambda > a$$

$$\theta = \frac{\pi}{\lambda} \times \frac{a \sin \theta}{\Delta x}$$

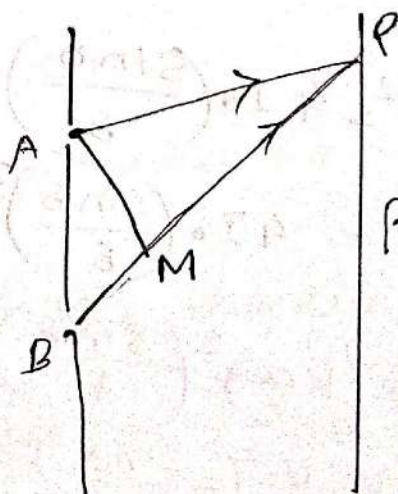
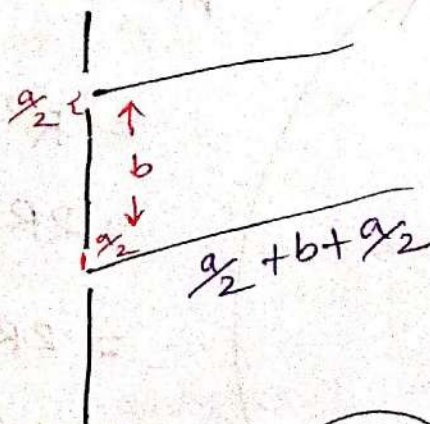
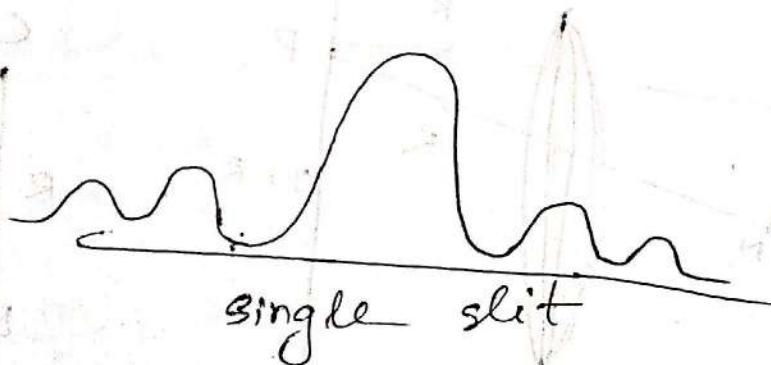
$$I_P = R^2 = I_0 \left(\frac{\sin \theta}{\theta} \right)^2$$

Diffraction Maxima

$$a \sin \theta = (2m+1) \frac{\lambda}{2}$$

Minima

$$a \sin \theta = n \lambda$$



$$I_P = 4 I_0 \cos^2 \frac{\beta}{2}$$

$$\beta = \frac{\pi}{\lambda} \times (a+b) \sin \theta$$

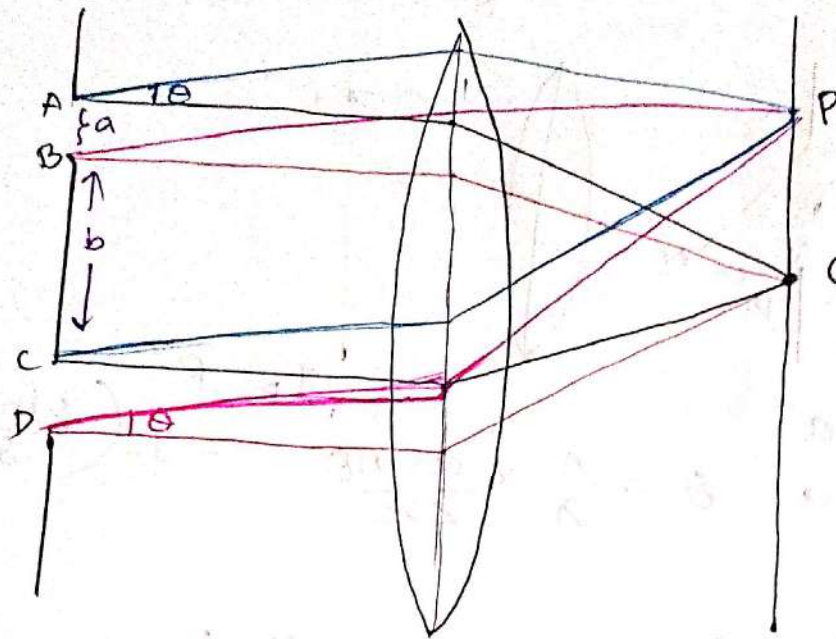
$$BM = (a+b) \sin \theta$$

Interference Maxima

$$(a+b) \sin \theta = m \lambda$$

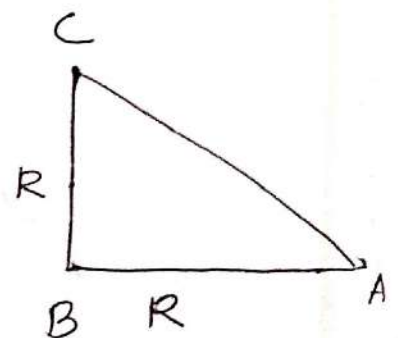
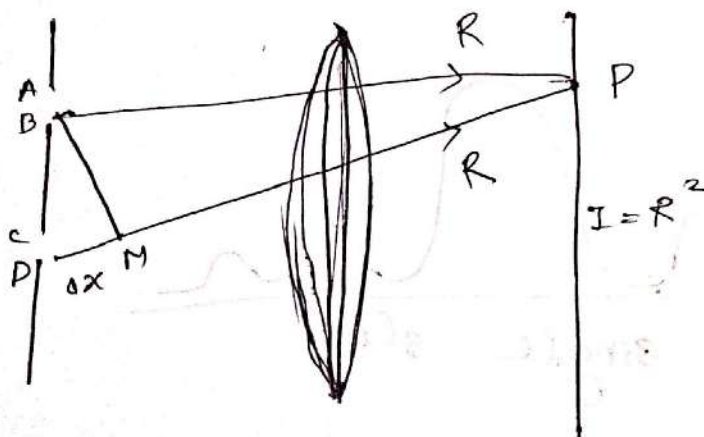
Minima

$$(a+b) \sin \theta = (2m+1) \frac{\lambda}{2}$$



B ও A corresponding
line 2 (নিত D)

O [central maxima]
[path difference
0 (Zero)]



$$AC^2 = 4 I_0 \left(\frac{\sin \delta}{\delta} \right)^2 \cos^2 \beta$$

$$I = 4 I_0 \left(\frac{\sin \delta}{\delta} \right)^2 \cos^2 \beta$$

$$R^2 = (ka)^2 \left(\frac{\sin \delta}{\delta} \right)^2$$

$$= a_0^2 \left(\frac{\sin \delta}{\delta} \right)^2$$

$$= I_0 \left(\frac{\sin \delta}{\delta} \right)^2$$

$$AC^2 = AB^2 + BC^2 + 2ABBC \cos \delta$$

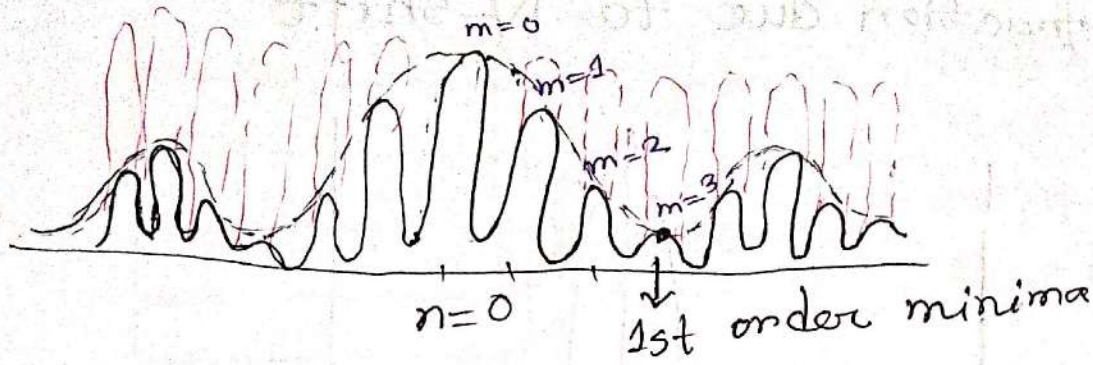
$$= R^2 + R^2 + 2R^2 \cos \delta$$

$$= 2R^2 + 2R^2 \cos \delta$$

$$= 2R^2 (1 + \cos \delta)$$

$$= 2R^2 \times 2 \cos^2 \delta/2$$

$$AC^2 = 4R^2 \cos^2 \beta$$



Missing Order

$$a \sin \theta = n \lambda \quad \rightarrow \text{দুইটি কাত- (effect) করছে}$$

$$(a+b) \sin \theta = m \lambda$$

$$\frac{a+b}{a} = \frac{m}{n} \quad [2a = b \rightarrow \text{যদি হয়}]$$

$$\Rightarrow \frac{a+2a}{a} = \frac{m}{n}$$

$$\Rightarrow \frac{3a}{a} = \frac{m}{n}$$

$$\Rightarrow \frac{m}{n} = 3$$

$$\Rightarrow m = 3n$$

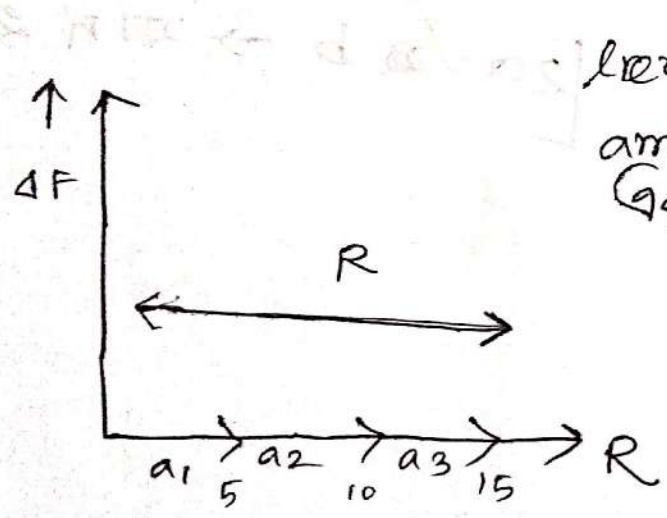
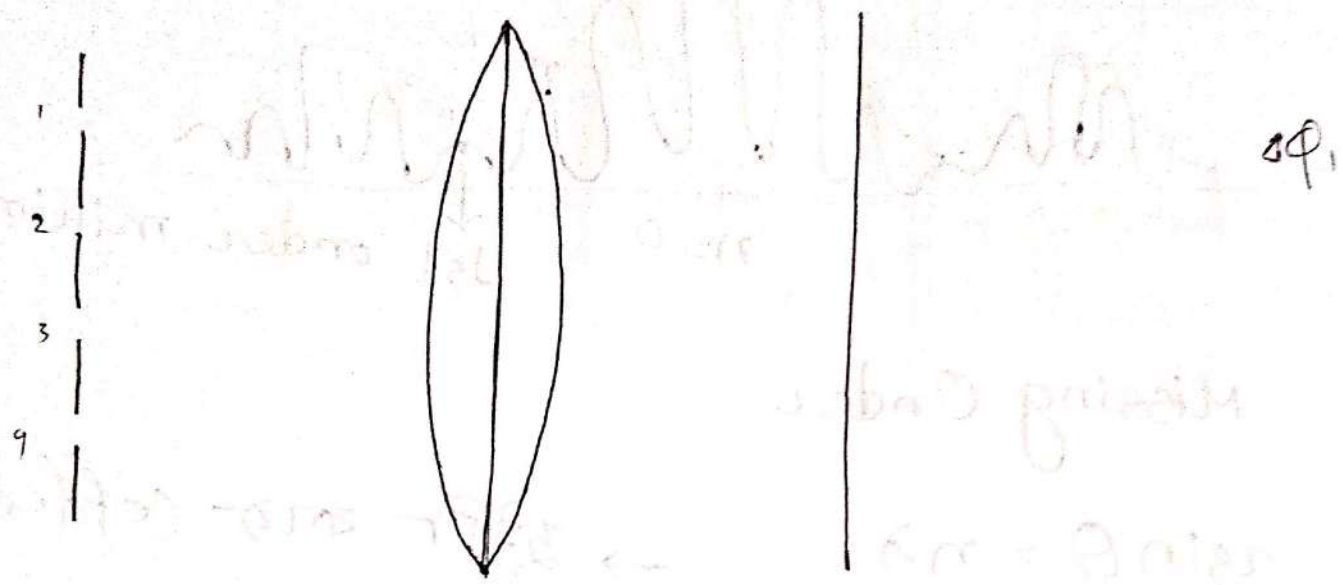
$\hookrightarrow n = 1, 2, 3$ (...th secondary minima)
 $m = 3, 6, 9$ (...th secondary maxima)

1st order minima $\rightarrow$ 3rd order maxima

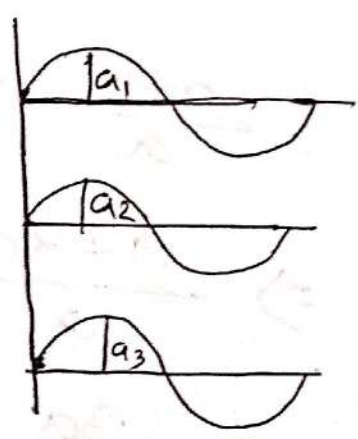
3, 6, 9 $\rightarrow$ missing হবে

$$\left[\begin{array}{l} a=b \text{ হলে } 2n=m \\ \text{হবে} \end{array} \right]$$

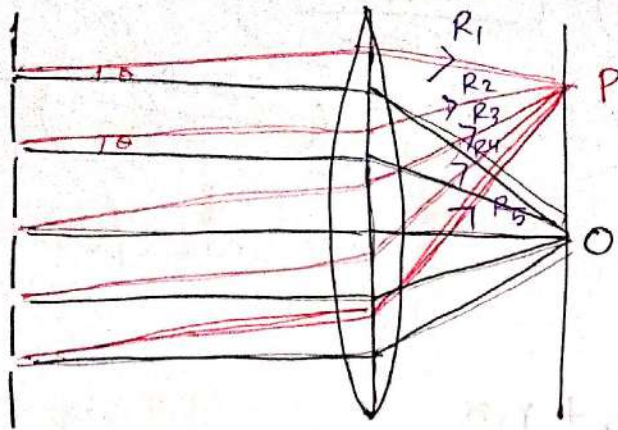
Diffraction due to N-shift



length হবে
amplitude
এর সমান

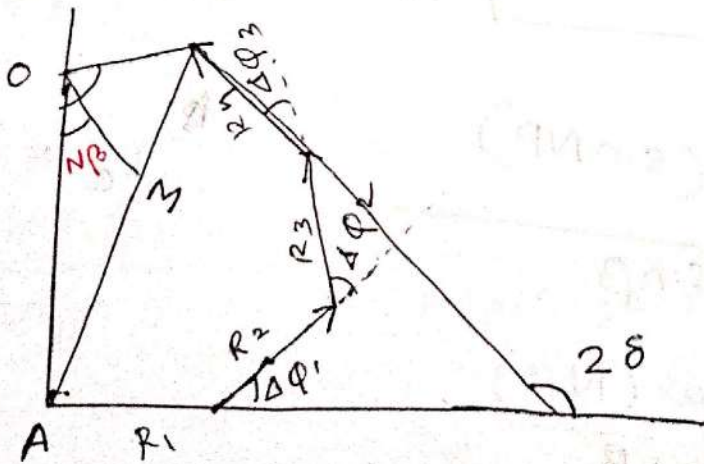
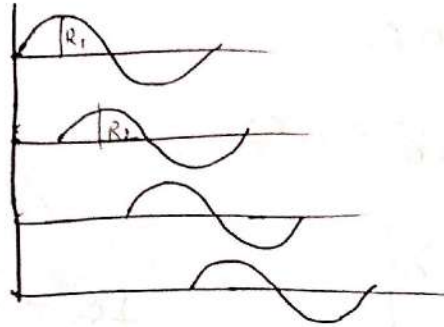


লুয়া



N স্লিট slit
থেকে secondary
wave resultant
থাকে N স্লিট

Vector policole Method



$$\sin N\beta = \frac{AB}{OA}$$

$$AB = OA \sin(N\beta)$$

$$2 OA = \frac{R}{\sin \beta}$$

$$\begin{aligned} AB &= 2 AB \\ A &= 2 \cdot OA \sin(N\beta) \\ &= R \frac{\sin N\beta}{\sin \beta} \end{aligned}$$

↑ diffraction term

$$A = (ka) \left(\frac{\sin \delta}{\delta} \right) \left(\frac{\sin N\beta}{\sin \beta} \right)$$

$$I_p = A^2 = I_0 \left(\frac{\sin \delta}{\delta} \right)^2 \left(\frac{\sin N\beta}{\sin \beta} \right)^2$$

↓ interference term

$$\sin \beta = 0$$

$$\sin \beta = 0 = \sin \pm n\pi$$

$$\beta = \pm n\pi \quad n=0, \beta=0$$

$$\beta = \frac{\pi}{\lambda} \times (a+b) \sin \theta$$

$$n\pi = \frac{\pi}{\lambda} \times (a+b) \sin \theta$$

$$\boxed{(a+b) \sin \theta = n\lambda}$$

$$I_c = I_0 \left(\frac{\sin \delta}{\delta} \right)^2 N^2$$

$$I_c = I_0 N^2$$

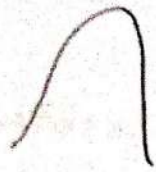
$$\lim_{n\pi \rightarrow 0} \frac{\frac{d}{d\beta} (\sin N\beta)}{\sin \beta}$$

$$= \frac{N \cos(N\beta)}{\cos \beta}$$

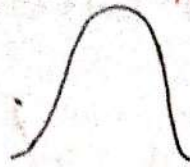
$$= \pm N$$

central maxima

1st order maxima



$n=0$



$n=1$

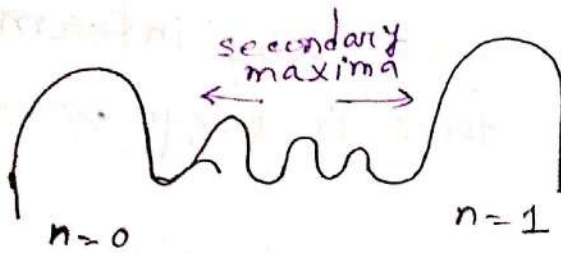
$$\sin N\beta = 0$$

$$N\beta = \pm K\pi$$

$$\beta = \pm \frac{K}{N} \pi$$

$$K \neq 0$$

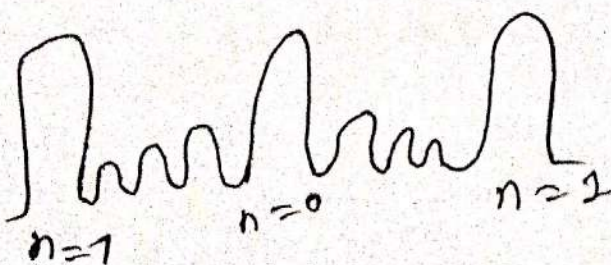
$$K = 1, 2, 3, \dots, (N-1)$$



$$\frac{\sin N\beta}{\sin \beta} = \frac{N^2}{1 + (N^2 - 1) \sin^2 \beta}$$

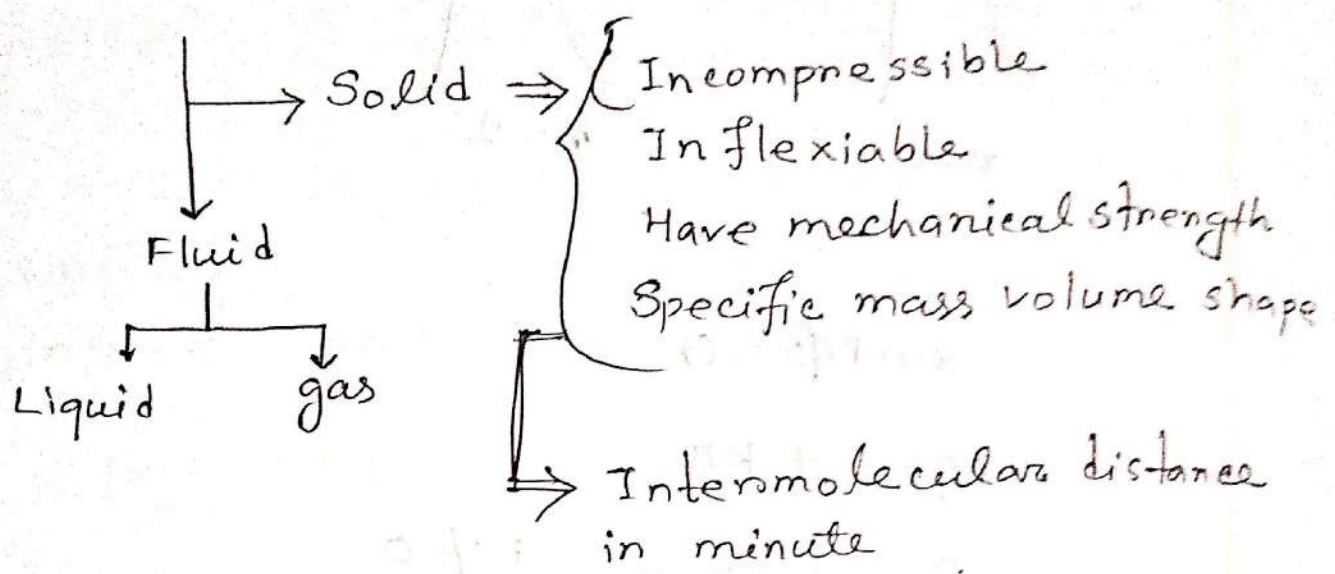
$$I = \frac{I_c}{1 + (N^2 - 1) \sin^2 \beta}$$

$$I_c = I_0 N^2$$

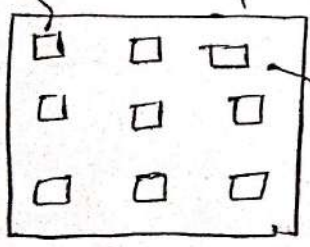


N slit এর জন্য

Structure of Matter



Desk/Lattice point
Regular and periodic
arrangement of
points in
space



student/
Basis

atom or group of
atom

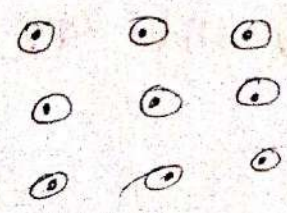
Room/space

That's why intermolecular
force is large.



Desk + student → classroom

Lattice point + Basis = crystal structure

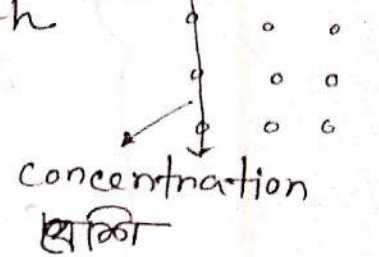
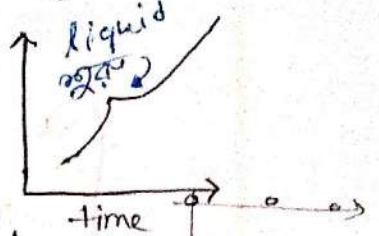
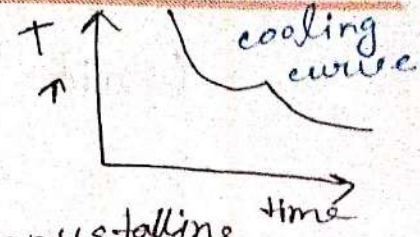


Solid

crystalline solid

Non-crystalline (Amorphous)

Poly crystalline



The constituents (atoms or molecules) are arranged in an orderly manner with definite geometric shape.

1) Isotropic → direction dependent

1) Symmetrical ⇒ Fixed melting point

Non-crystalline (Amorphous)

⇒ asymmetrical

⇒ Anisotropic → direction independent

Poly crystalline → Property show করে crystalline solid এর মতো

⇒ Powder resemble amorphous but properties are just like crystalline solid.

Unit Cell

⇒ is smallest repeating unit that form the total symmetry of the crystal.

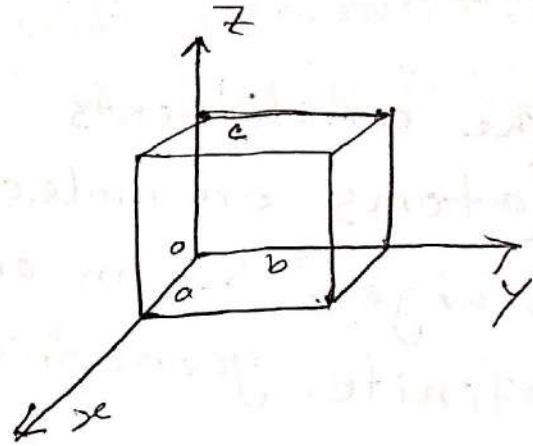
a, b, c

↓
axial length

α, β, γ

↓
axial angle

Lattice Parameters



7 - crystal system

Triclinic → $a \neq b \neq c$
 $\alpha \neq \beta \neq \gamma$

Monoclinic

Orthorhombic

Cubic → $a = b = c$
 $\alpha = \beta = \gamma = 90^\circ$

Trigonal

Tetragonal

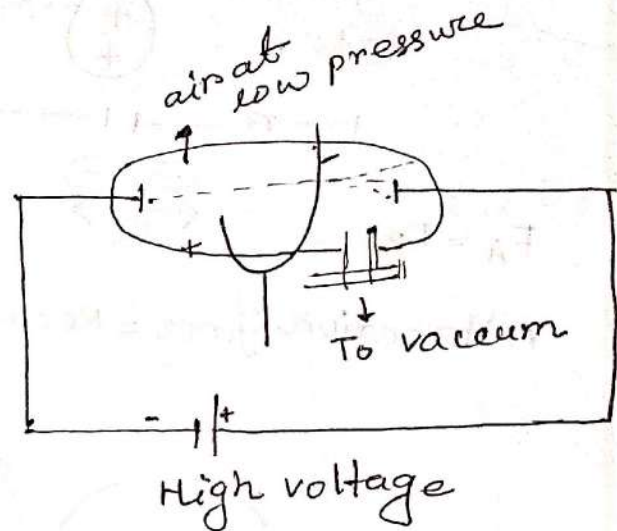
Hexagonal → $a = b \neq c$
 $\alpha = \beta = 90^\circ, \gamma = 120^\circ$

Atomic Model

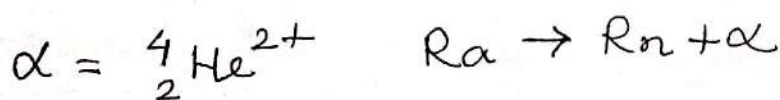
J.J Thompson $\rightarrow$ 1897 $\rightarrow$ Cathode ray experiment
 $\rightarrow$ 1904
plum-pudding Model

$\Rightarrow$ ক্যাথোড ray negatively charged

$\Rightarrow$ atom এর tiny particle আছে যা negatively charged অবস্থায় থাকে

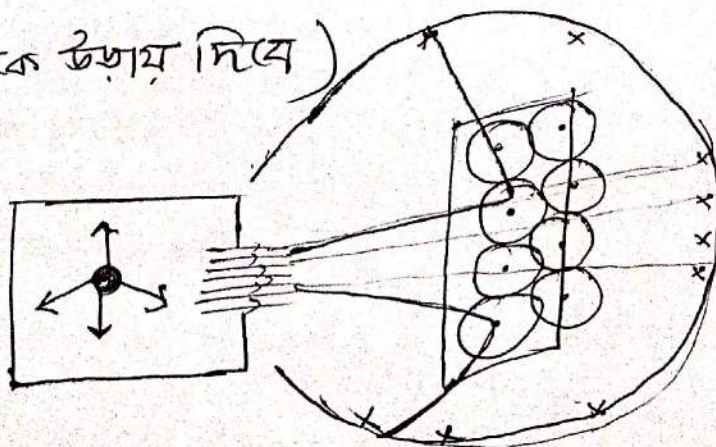


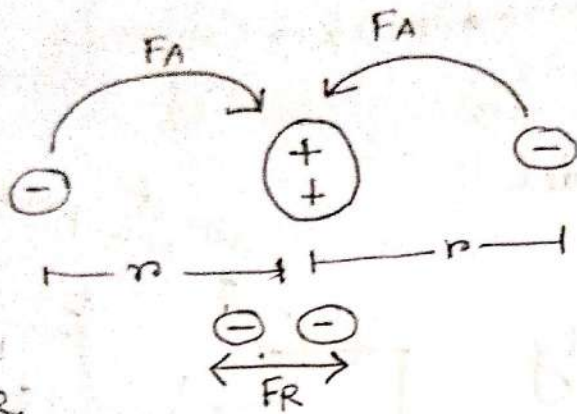
$\Rightarrow$ Atom is regarded as a sphere of uniformly distributed positive charge in which the electrons are embedded to make it electrically charge neutral.



$\rightarrow + \Rightarrow$ low angle scattering
 α -particle

(+) কে ঠোঁট দিয়ে)





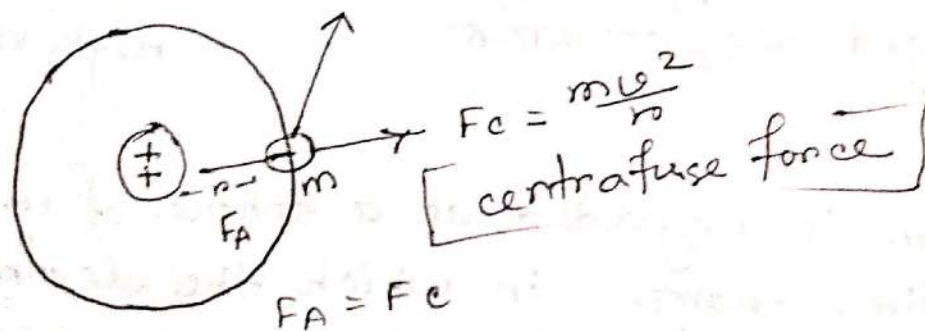
$$F_A = F_R$$

Attraction force = Repulsion force

$$F_A = \frac{2e \times e}{4\pi\epsilon_0 r^2}$$

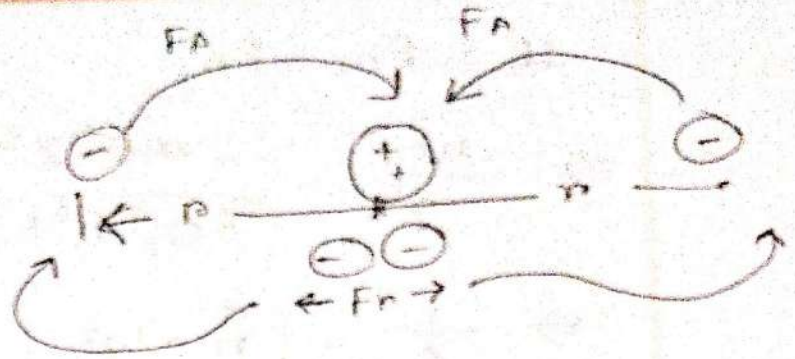
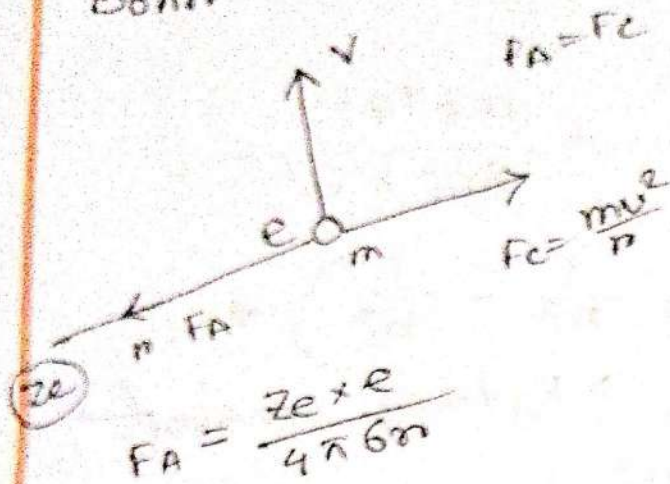
$$F_R = \frac{e \times e}{4\pi\epsilon_0 (2r)^2}$$

$$F_A > F_R \quad \swarrow \quad 8 \text{ times}$$

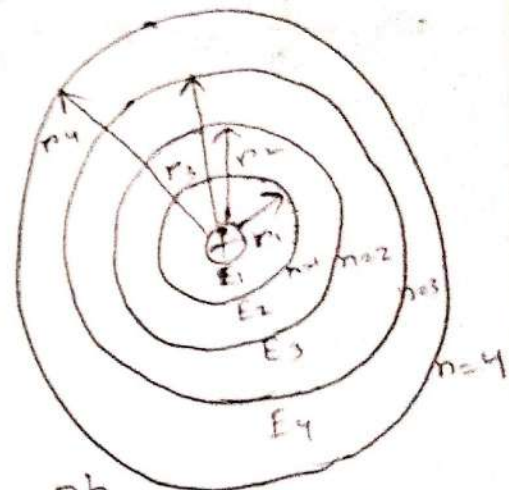


$$F_A = F_c$$

Bohr



$$E_1 < E_2 < E_3 < E_4$$



$$mvr = \frac{nh}{2\pi} \quad \text{--- (I)}$$

$$h = 6.63 \times 10^{-34}$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

$$m = 9.1 \times 10^{-31}$$

$$V = \frac{nh}{2\pi mr}$$

$$r = \frac{n^2 h^2 \epsilon_0}{\pi m z e^2}$$

$$r_n = 0.53 \times n^2 \text{ \AA}$$

$$K.E = \frac{1}{2}mv^2$$

$$P.E = q \times (-e)$$

$$= \frac{-ze^2}{4\pi\epsilon_0 r}$$

$$E_1 = -13.6 \text{ eV}$$

$$E_2 = -3.4 \text{ eV}$$

$$E_3 = -$$

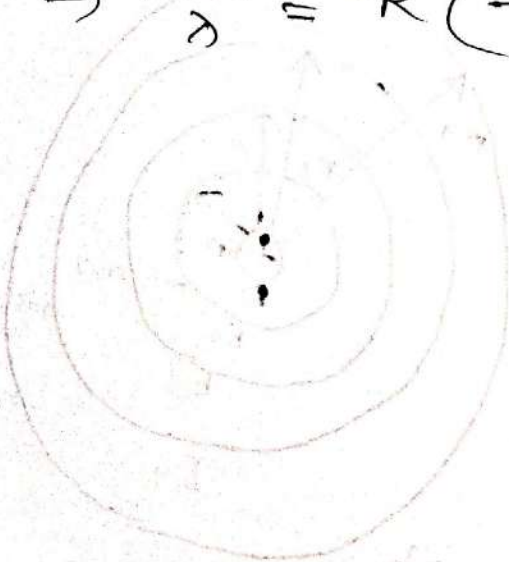
$$E_4 = -0.85 \text{ eV}$$

$$\Delta E = E_{n_2} - E_{n_1}$$

$$\Rightarrow \frac{hc}{\lambda} = \frac{-me^4 z^2}{8n_1^2 h^2 \epsilon_0^2} + \frac{me^4 z^2}{8n_2^2 h^2 \epsilon_0^2}$$

$$\Rightarrow \frac{1}{\lambda} = \left[\frac{me^4 z^2}{8h^2 \epsilon_0^2 c} \right] \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \rightarrow$$

$$\Rightarrow \frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ Rydberg constant}$$



(iii)

$$\frac{n_2}{n_1} = \text{constant}$$

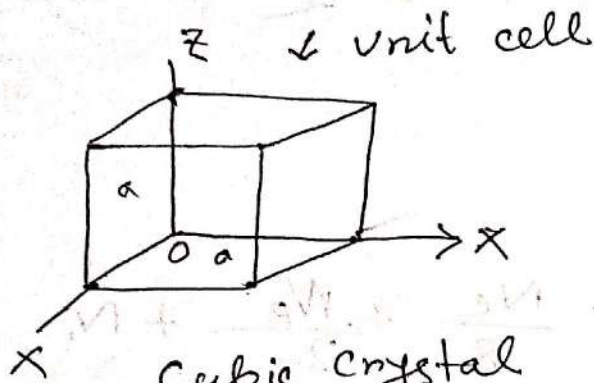
$$K.E = \text{Kinetic Energy}$$

$$P.E = \text{Potential Energy}$$

$$E = \text{Total Energy}$$

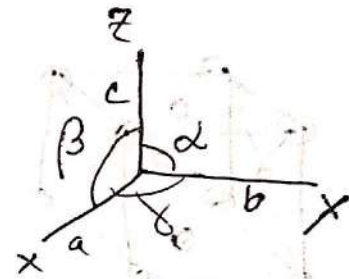
Packing Factor

↳ Packing factor indicates how atoms are closely packed in a unit cell.



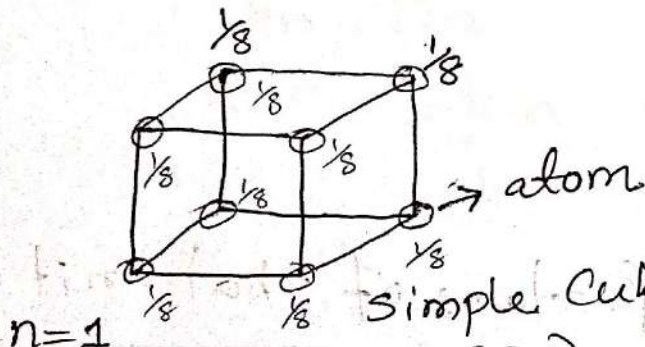
$$a = b = c$$

$$\alpha = \beta = \gamma = 90^\circ$$



Volume of unit cell = a^3

$$\frac{4}{3} \pi r^3$$



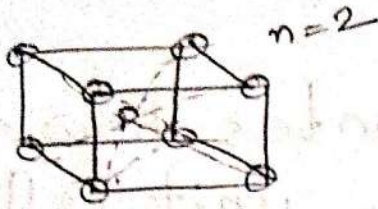
$n=1$

Simple Cubic → number of atom
(Sc) 2(7. 1

↗ centre

$$N = \underbrace{\frac{N_c}{8}}_{\text{corner}} + \underbrace{\frac{N_p}{2}}_{\text{surface}} + N_i \text{ (N inside)}$$

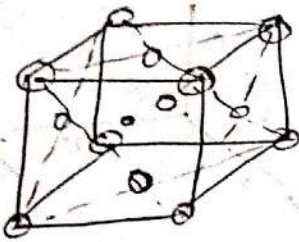
$$= \frac{8}{8} + 0 + 0 = 1 \rightarrow \text{number of lattice point} = \text{number of atom} \text{ (N)}$$



$$N = \frac{N_c}{8} + \frac{N_p}{2} + N_i$$

$$= \frac{8}{8} + 0 + 1$$

Body Centred Cubic
(BCC) = 2



$$N = \frac{N_c}{8} + \frac{N_p}{2} + N_i$$

$$= \frac{8}{8} + \frac{6}{2} + 0$$

$$= 1 + 3 = 4$$

$n=4$

$N=4$

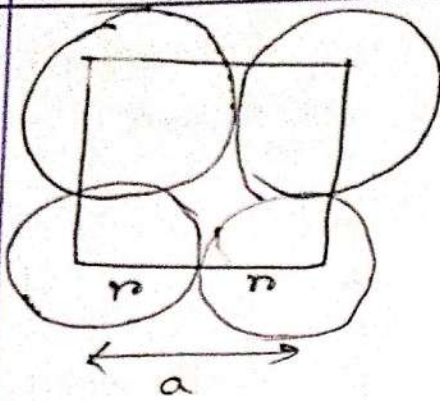
Face-centred
cubic (FCC)

Packing factor,

$$P.F = \frac{\text{Volume of atoms of that unit cell}}{\text{Volume of unit cell}}$$

$$= \frac{\text{no. of atom} \times \text{Volume of one atom}}{a^3}$$

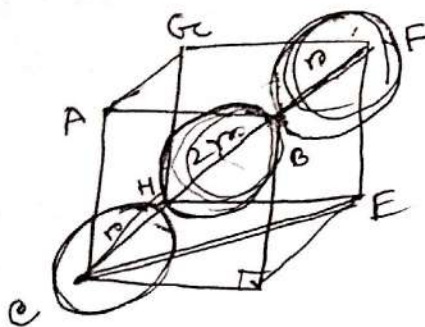
$$P.F = \frac{n \times \frac{4}{3} \pi r^3}{a^3} \times 100\%$$



$$a = r + r = 2r \Rightarrow r = \frac{a}{2}$$

$$P.F = 52\%$$

$$\begin{aligned} \text{Empty Space} &= (100 - 52)\% \\ &= 48\% \end{aligned}$$



$$\triangle CEF \sim \triangle$$

$$\begin{aligned} CF &= r + 2r + r \\ &= 4r \end{aligned}$$

$$CF = \sqrt{3}a$$

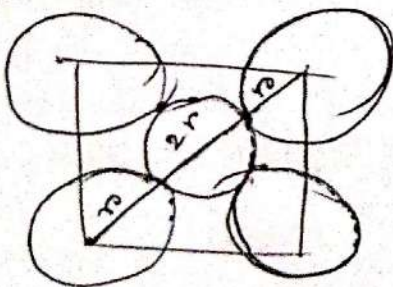
$$\begin{aligned} CF^2 &= CE^2 + EF^2 \\ &= CE^2 + a^2 \\ &= 2a^2 + a^2 \\ &= 3a^2 \end{aligned}$$

$$\begin{aligned} \therefore 4r &= \sqrt{3}a \\ \Rightarrow r &= \frac{\sqrt{3}}{4}a \end{aligned}$$

$$n = 2$$

$$\begin{aligned} \text{Empty space} &= (100 - 68)\% \\ &= 32\% \end{aligned}$$

$$P.F = 68\%$$



$$\begin{aligned} AC &= r + 2r + r \\ &= 4r \end{aligned}$$

$$AC = \sqrt{2}a$$

$$r = \frac{\sqrt{2}}{4}a$$

$$n = 4$$

$$\triangle ABC$$

$$\begin{aligned} AC^2 &= AB^2 + BC^2 \\ &= 2a^2 \end{aligned}$$

$$AC = \sqrt{2}a$$

$$P.F = 74\%$$

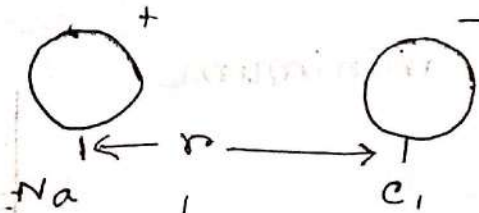
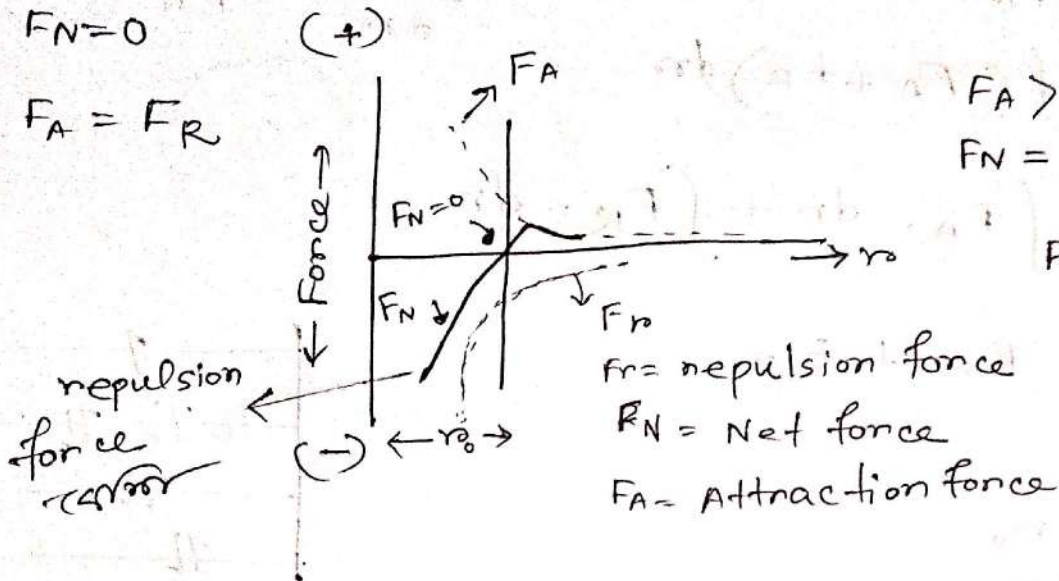
$$\text{Empty space} = (100 - 74)\% = 26\%$$

$r = r_0$
 $F_N = 0$
 $F_A = F_R$

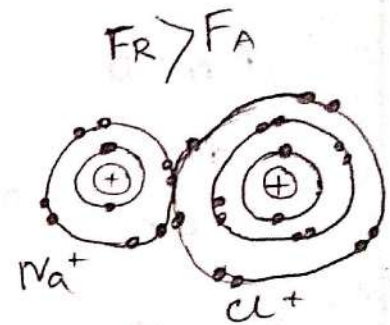
$F_N = F_A + F_R$

$F_A > F_R$
 $F_N = F_A (+)$

$F_R > F_A$
 $F_N = F_R (-)$



interatomic separation



$F_A = \text{negligible}$

Metallic Bond $\rightarrow$ unsaturated Bond

Atomic $\rightarrow$ AAM

Buffer effect $\rightarrow$ MNZ

$$E_T = \int F_N dr$$

$$= \int (F_A + F_R) dr$$

$$= \int F_A \cdot dr + \int F_R \cdot dr$$

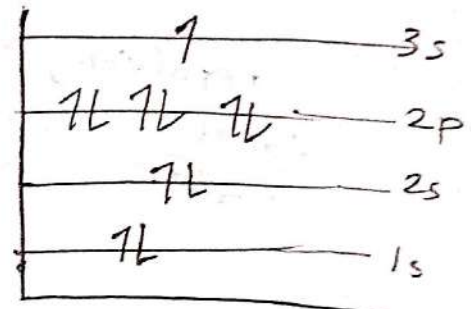
$$E_T = E_A + E_R$$

$$r = r_0$$

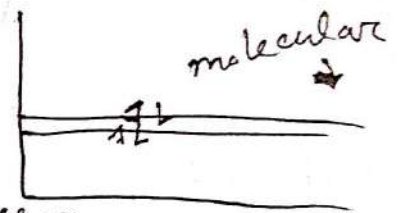
$$F_N = 0$$

$$\frac{dE_T}{dr} = 0$$

$$E_T = \text{minimum}$$



Na

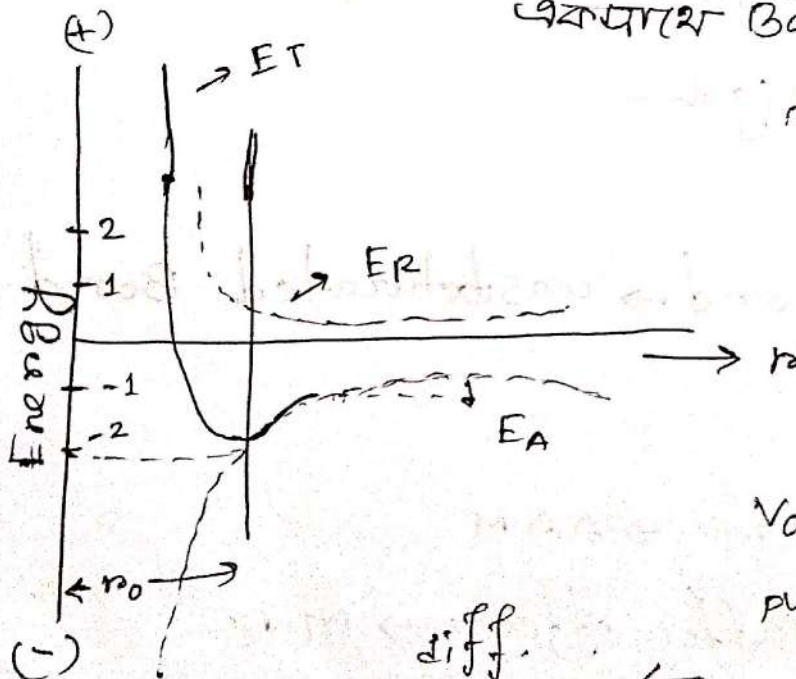


molecular

আনেকগুলো
একসাথে Valence Band তৈরি করে।

$r_0 \rightarrow$ energy minimum

Valence Band.
conduction band
energy gap

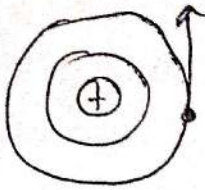


diff.
energy state
এ শরৎ

Valence Band $\rightarrow e^-$ শরৎ
push apply করে
conduction band
শরৎ।

Matter waves

$$mvr = \frac{nh}{2\pi}$$



e
 m
 E

non-radiating stable orbit

$$r = \frac{n^2 h^2 \epsilon_0}{\pi m z e}$$

$$E = h\nu$$

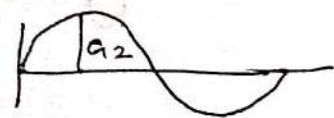
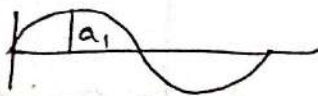
$$p = \frac{h\nu}{c}$$

$$r \propto n^2$$

$$p = \frac{h}{\lambda}$$

$$c = \nu \lambda$$

$$\frac{\nu}{c} = \frac{1}{\lambda}$$



wavelength of matter wave

$$\lambda = \frac{h}{mv}$$

mass of matter particle

$$A = a_1 \neq a_2$$

$$I \propto A^2$$

de-Broglie wavelength

$$\# \frac{1}{2} mv^2 = eV$$

$\Rightarrow$

$$mv^2 = 2eV$$

$$m^2 v^2 = 2meV$$

$$mv = \sqrt{2meV}$$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

matter wave $\rightarrow$ not electro magnetic wave

$$2\pi r = n\lambda$$

$$\lambda = \frac{h}{mv}$$

$$2\pi r = \frac{nh}{mv}$$

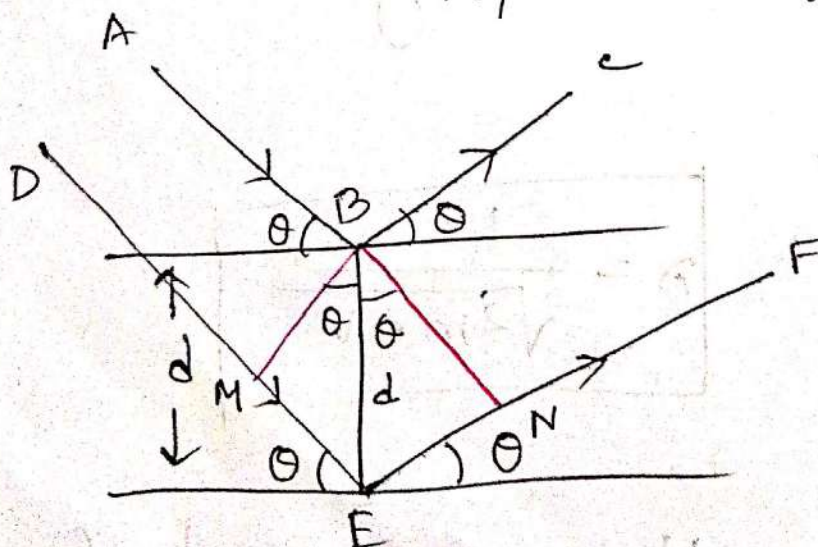
Matter wave

i) New kind of wave

ii)

iii) Pilot. wave their function is to guide the matter particle.

Davission - German
Experiment.



Bragg's law

$$\Delta x = ME - EN$$

↓
Path
difference

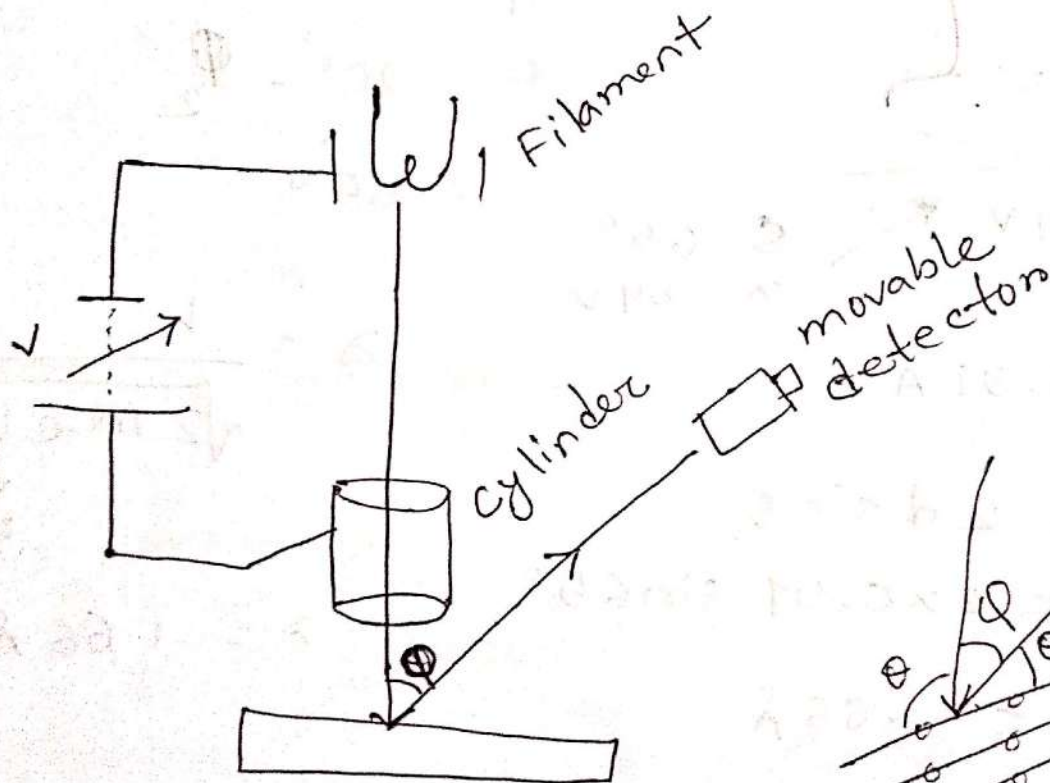
$$42e = ME + EN$$

$$= 2ME$$

$$= 2d \sin \theta$$

$$ME = d \sin \theta$$

$$\Rightarrow \boxed{2d \sin \theta = n\lambda} \rightarrow \text{Bragg's law}$$



Target: Nickel crystal

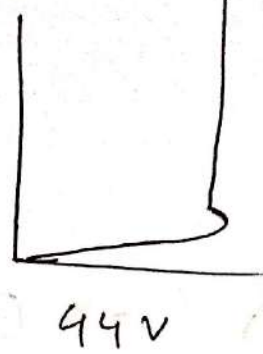
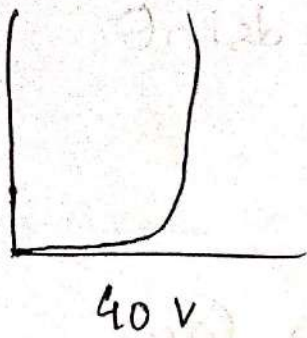
$$2d \sin \theta = \lambda$$

$$\Rightarrow \lambda = 2d \sin \theta$$

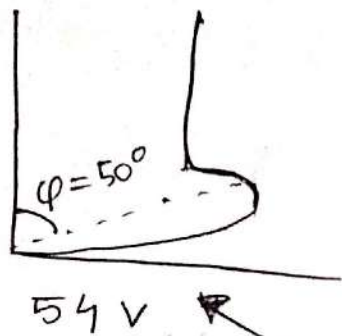
$$\theta + \theta + \phi = 180^\circ$$

$$\Rightarrow \theta = 90^\circ - \frac{\phi}{2}$$

$$\lambda = \frac{h}{\sqrt{2meV}}$$



voltage graph



$\theta = 65^\circ$
 $\lambda = 54 \text{ V}$

$\phi = 50^\circ$

$\theta = 90^\circ - \frac{\phi}{2}$
 $= 65^\circ$

$d = 0.91 \text{ \AA}$

$\lambda = \frac{h}{\sqrt{2meV}}$
 $\downarrow$
 54 V

$a = 2d \sin \theta$
 $= 2 \times 0.91 \sin 65^\circ$
 $= 1.65 \text{ \AA}$

$\lambda = 1.66 \text{ \AA}$

(proved) that, electron acts like wave

$$y = a \sin(\omega t - kx)$$

$$\omega t - kx = \text{const}$$

$$\Rightarrow \omega dt - k dx = 0$$

$$\Rightarrow \omega dt = k dx$$

$$\omega = 2\pi n$$

$$k = \frac{2\pi}{\lambda}$$

$$v_p = \frac{dx}{dt} = \frac{\omega}{k} = n\lambda$$

phase velocity or
wave velocity

#

$$v_p = \nu \lambda$$

$$\Rightarrow \lambda = \frac{v_p}{\nu} \quad \text{--- (1)}$$

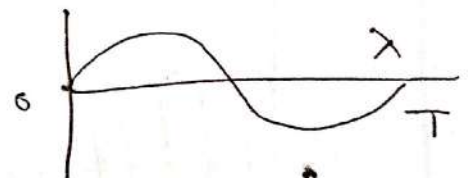
$$E = mc^2 = h\nu$$

$$m = \frac{h\nu}{c^2} \quad \text{--- (11)}$$

$$\lambda = \frac{h}{m\nu}$$

$$\frac{v_p}{\nu} = \frac{h}{\frac{h\nu}{c^2}}$$

$$\Rightarrow v_p = \frac{c^2}{\nu}$$



$$v_p = \frac{\lambda}{T}$$

$$v_p = n\lambda$$

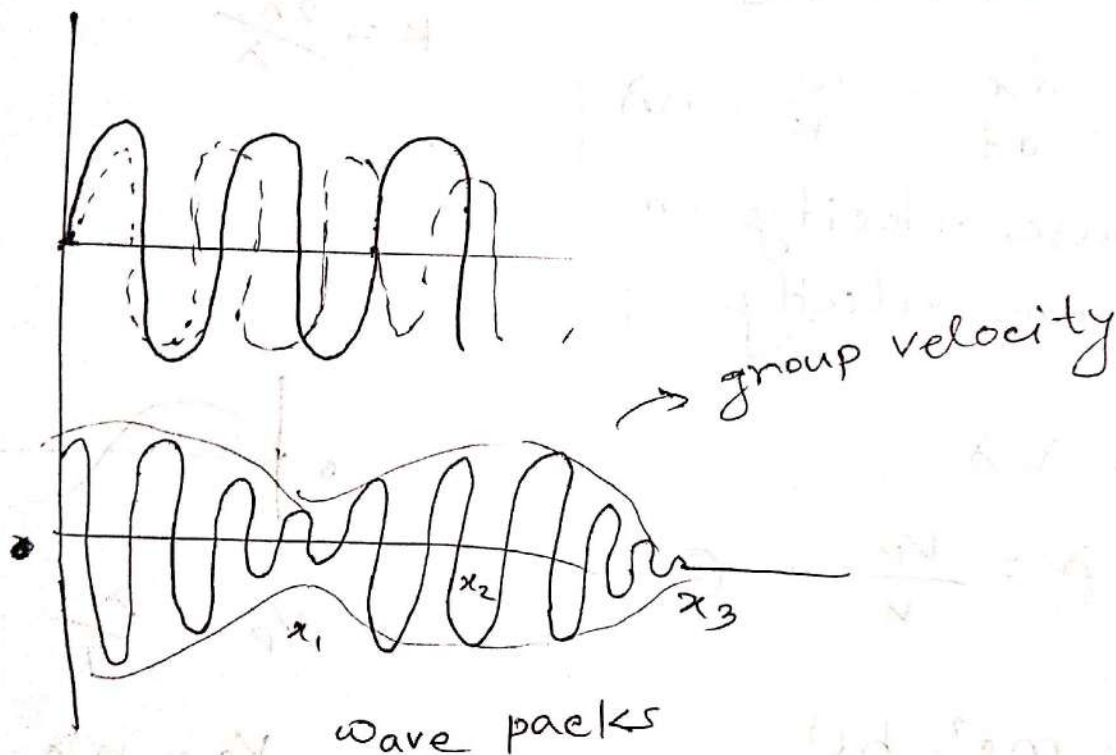
↓
phase velocity

group velocity

group velocity

$$y_1 = a \sin(\omega_1 t - k_1 x)$$

$$y_2 = a \sin(\omega_2 t - k_2 x)$$



~~y~~ =

$$Y = Y_1 + Y_2$$

$$= a \sin(\omega_1 t - k_1 x) + a \sin(\omega_2 t - k_2 x)$$

$$Y = 2 \cos\left(\frac{1}{2} \Delta \omega t - \frac{1}{2} \Delta K x\right) a \sin(\omega t - kx)$$

$\downarrow$
 constant

$$\frac{1}{2} \Delta \omega t - \frac{1}{2} \Delta k x = \text{const.}$$

$$\Rightarrow \Delta \omega dt - \Delta k dx = 0$$

$$v_g \Rightarrow \frac{dx}{dt} = \frac{\Delta \omega}{\Delta k}$$

$$v_p = \frac{\omega}{k}$$

$$\Rightarrow \omega = v_p k$$

$$v_g = \frac{\Delta (v_p k)}{\Delta k}$$

$$\Rightarrow v_g = k \frac{\Delta v_p}{\Delta k} + v_p$$

$$v_g = v_p + k \frac{\Delta v_p}{\Delta k}$$

$$k = \frac{2\pi}{\lambda}$$

$$\Rightarrow \frac{\Delta k}{\Delta \lambda} = -\frac{2\pi}{\lambda^2}$$

$$\Rightarrow \Delta k = -\frac{2\pi}{\lambda^2} \Delta \lambda$$

$$\Delta k = -\frac{2\pi}{\lambda^2} \Delta \lambda$$

$$\therefore v_g = v_p + \frac{2\pi}{\lambda} \frac{\Delta v_p}{-\frac{2\pi}{\lambda^2} \Delta \lambda}$$

$$v_g = v_p - \lambda \frac{\Delta v_p}{\Delta \lambda}$$

Dispersive medium $\therefore$
 $v_g \neq v_p$

$$\left[\begin{array}{l} \text{Dispersive} \\ \frac{\Delta v_p}{\Delta \lambda} \neq 0 \end{array} \right]$$

Non-dispersive, $\frac{\Delta V_p}{\Delta \lambda} = 0$

$$V_g = V_p$$